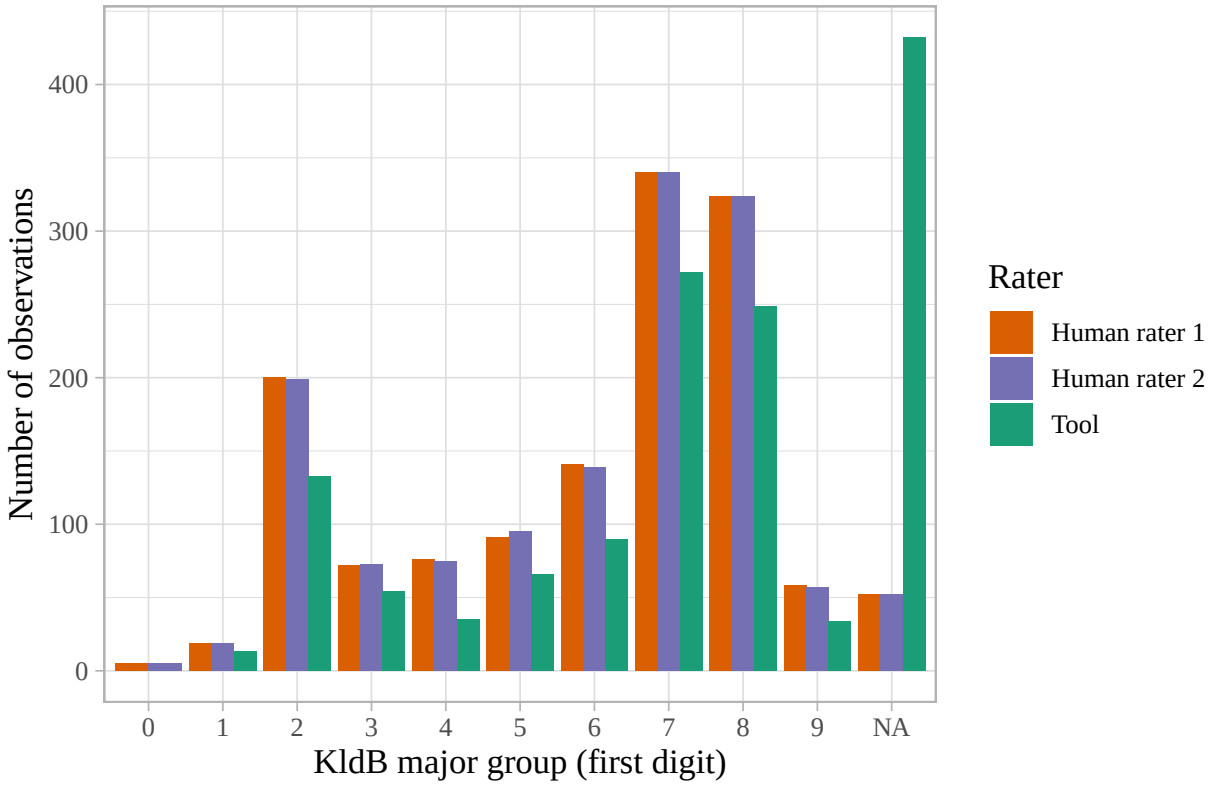
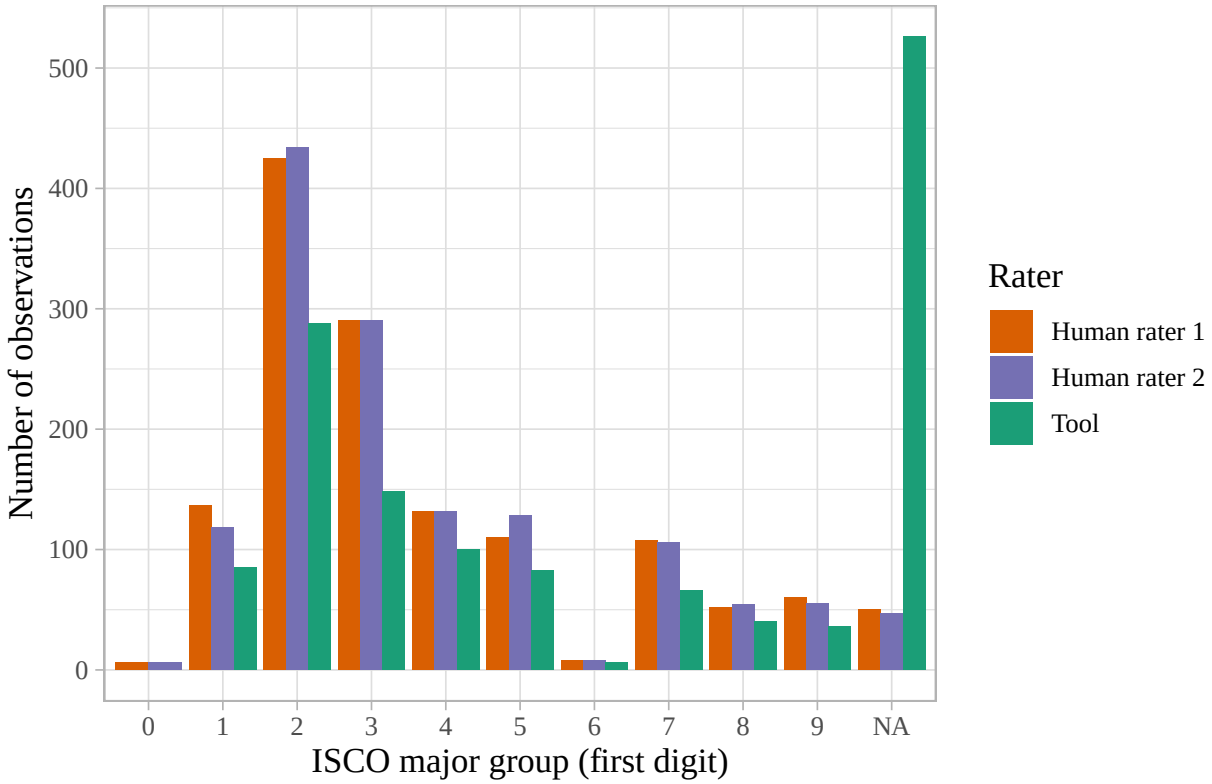
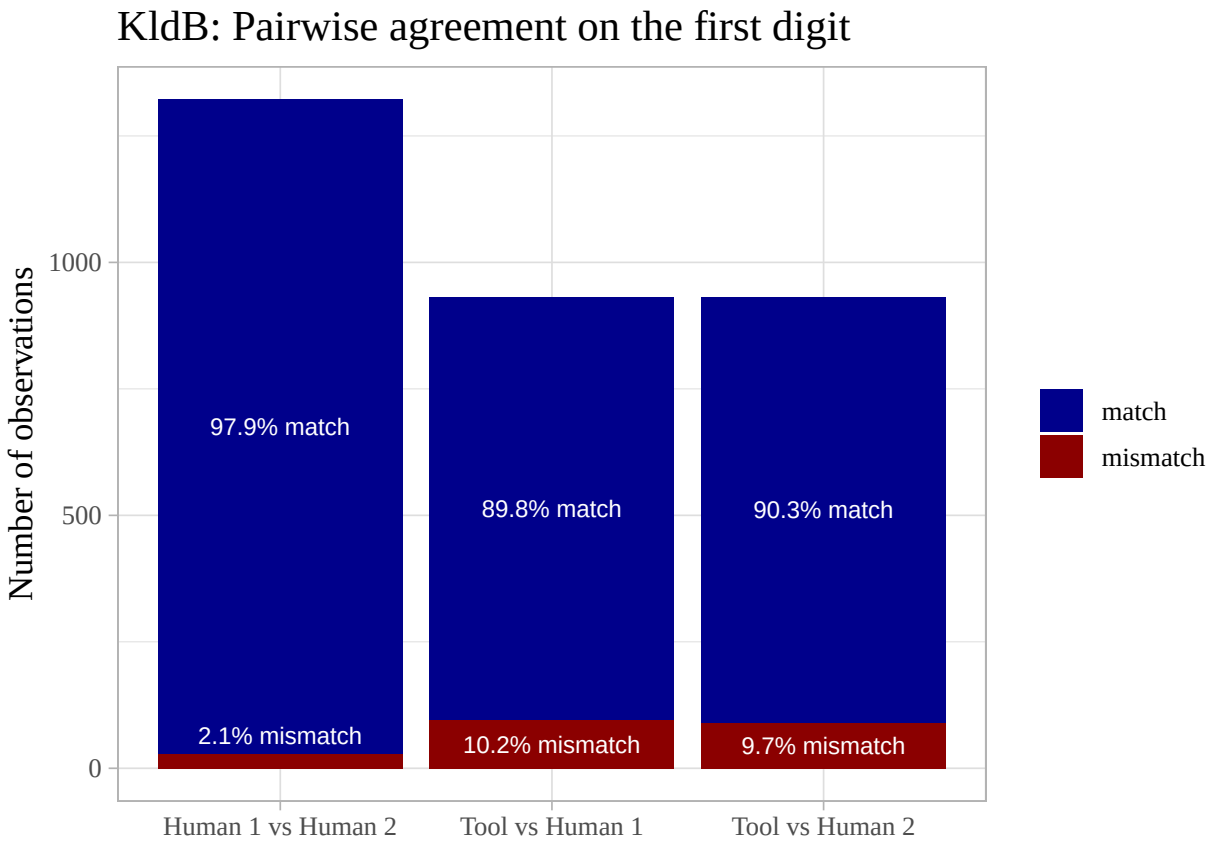
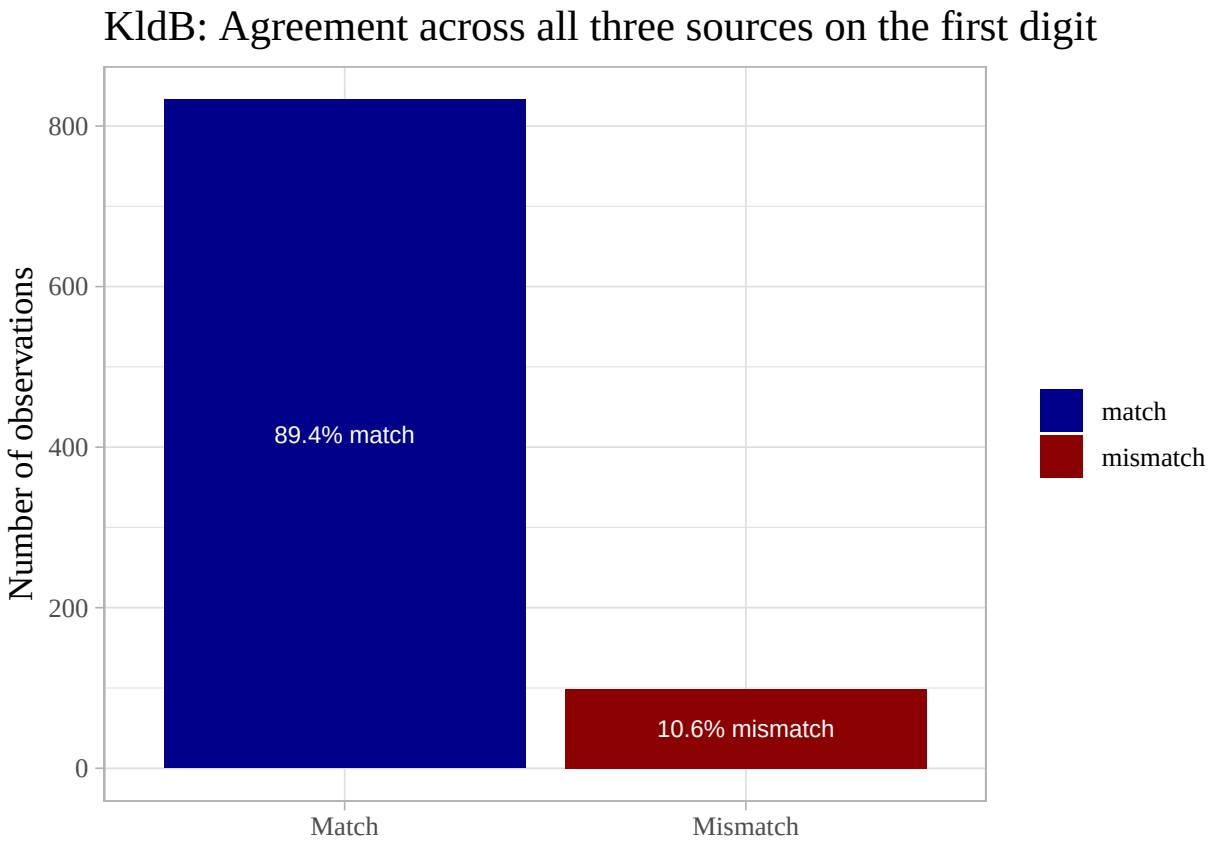
model_basis_isco_all

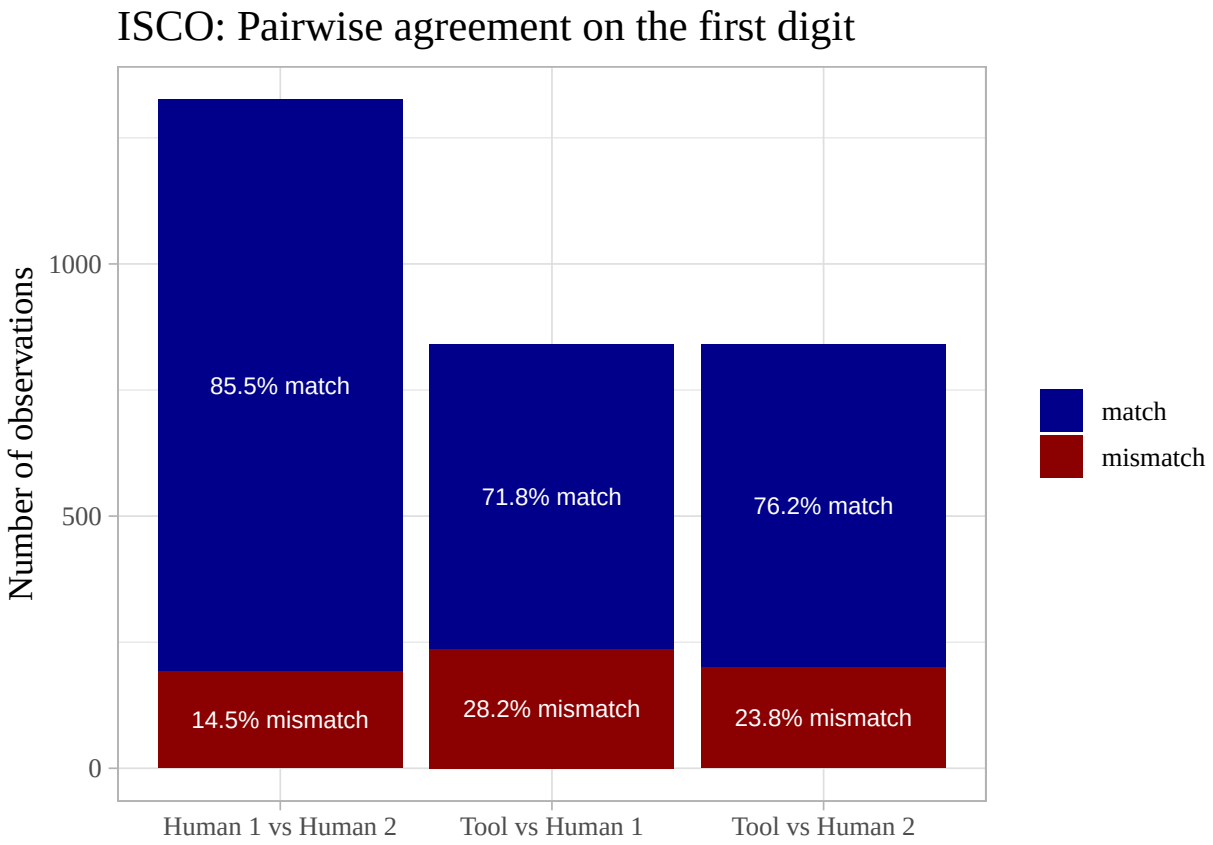
```
## [1] "C:/Users/jeroe/Uni_C/noisy_measurement/src/data_preparation.R"
```

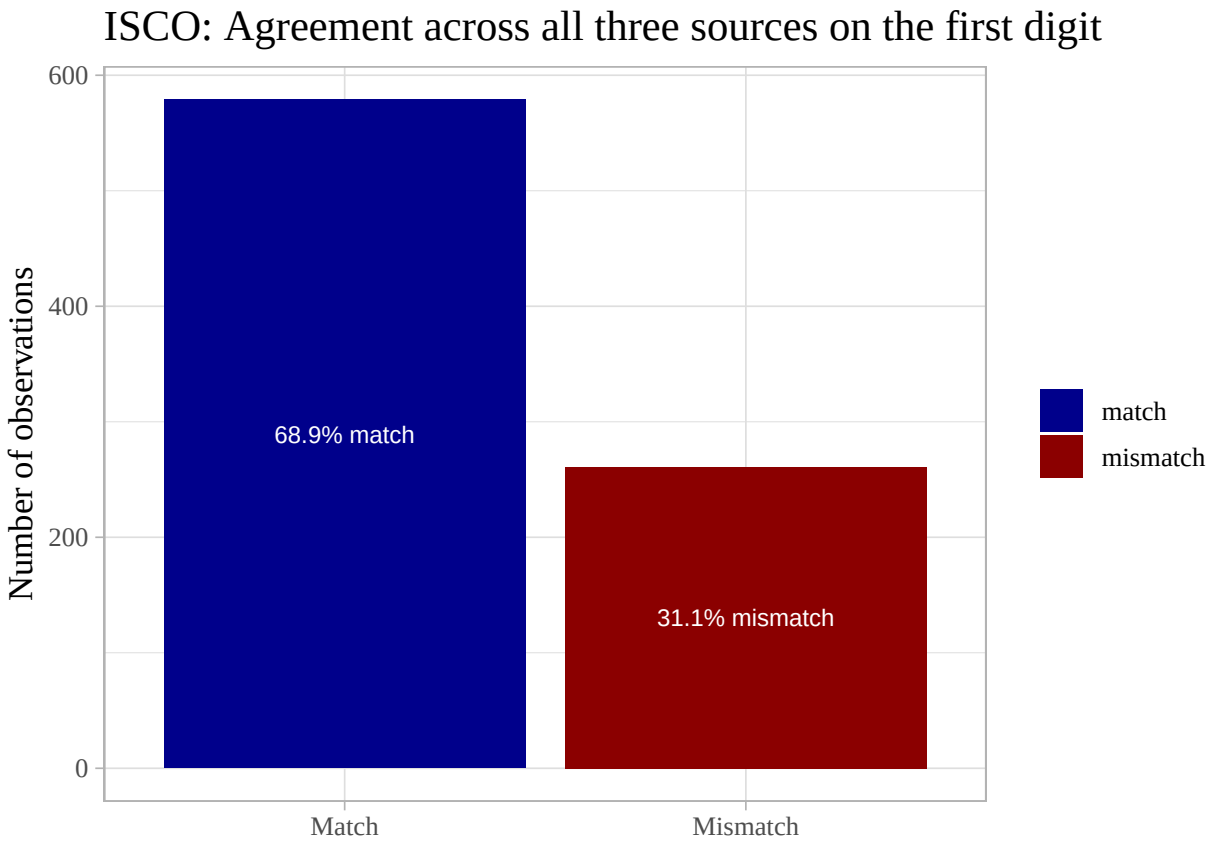


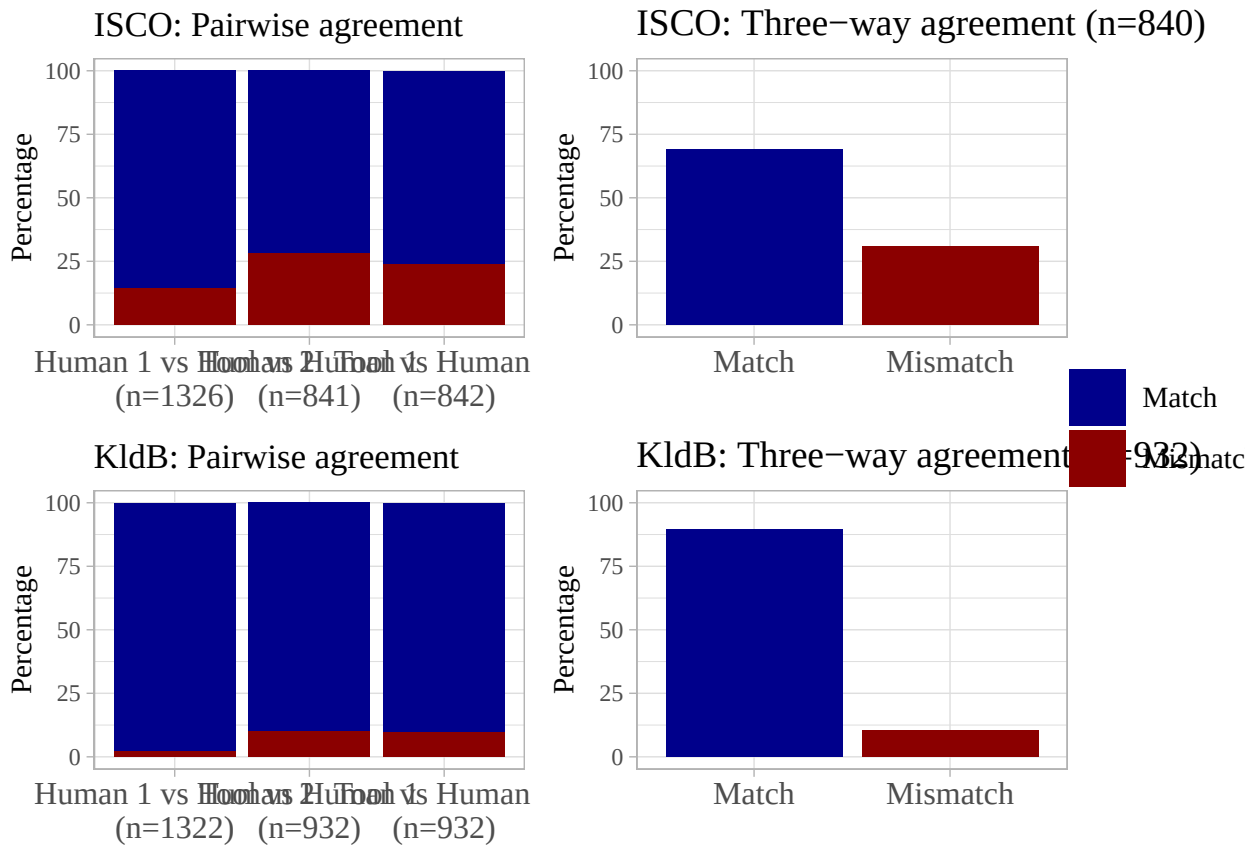

Number of observations
500
400
300
200
100
0
0
1
2
3
4
5
6
7
8
9
NA
ISCO major group (first digit)
Rater
Human rater 1
Human rater 2
Tool


KldB: Pairwise agreement on the first digit
Number of observations
1000
500
0
97.9% match
2.1% mismatch
89.8% match
10.2% mismatch
90.3% match
9.7% mismatch
Human 1 vs Human 2
Tool vs Human 1
Tool vs Human 2
match
mismatch


KldB: Agreement across all three sources on the first digit
Number of observations
800
600
400
200
0
89.4% match
10.6% mismatch
Match
Mismatch
match
mismatch


ISCO: Pairwise agreement on the first digit
Number of observations
1000
500
0
85.5% match
14.5% mismatch
71.8% match
28.2% mismatch
76.2% match
23.8% mismatch
Human 1 vs Human 2
Tool vs Human 1
Tool vs Human 2
match
mismatch


ISCO: Agreement across all three sources on the first digit
Number of observations
600
400
200
0
68.9% match
31.1% mismatch
Match
Mismatch
match
mismatch




```
# Packages
library(data.table)
library(dplyr)
library(tidyr)
library(cmdstanr)



# keep only relevant columns
isco_data <- data_basis %>%
  select(item_id, isco_tool_1st, res1_isco_1st, res2_isco_1st)

#Rename columns
setnames(
  data_basis,
  old = c("isco_tool_1st", "res1_isco_1st", "res2_isco_1st"),
  new = c("isco_tool", "isco_human1", "isco_human2")
)


head(data_basis)
```

```
##    kldb_tool_vorlaufig kldb_tool_mit_anfniveau res1_kldb res2_kldb isco_tool_vorlaufig isco_tool_mit
##                  <char>                   <char>    <char>    <char>              <char>
## 1:               81883                    81883     81883     81883                2433
## 2:               52112                    52112     52112     52112                8322
```

```
## 3:               <NA>                  <NA>     83131     83131               <NA>
## 4:              23411                 23411     23411     23411               9329
## 5:              72134                 72134     72132     72132               2412
## 6:              25104                 25104     27224     27224               2144
##    res1_isco res2_isco kldb_infas_num                                kldb_infas_str isco_infas_nu
##       <char>    <char>         <char>                                        <char>         <cha
## 1:      2433      2433          81883          [81883] Pharmazie (s.s.T.) - Spezialist           243
## 2:      8322      8322          52112   [52112] Berufskraftfahrer(Pers./PKW.)-Fachkraft           832
## 3:      3412      3412          83131 [83131] Heilerziehungspflege, Sonderpaed. -Helfer           341
## 4:      7322      7322          23412                  [23412] Drucktechnik - Fachkraft           732
## 5:      4312      4312          72132        [72132] Versicherungskaufleute - Fachkraft           431
## 6:      2149      2144          27224     [27224] Konstruktion und Geraetebau - Experte           214
##
##
## 1: [2433] Akademische und vergleichbare Fachkräfte im Bereich Vertrieb (Technik und Medizin, ohne Ini
## 2:                                                                [8322] Personenkraftwagen-, Ta
## 3:                                                                                [3412] Nicht
## 4:
## 5:                                                                 [4312] Bürokräfte in der Sta
## 6:
##    kldb_tool_1st res1_kldb_1st res2_kldb_1st isco_tool isco_human1 isco_human2 item_id
##           <char>        <char>        <char>    <char>      <char>      <char>   <int>
## 1:            8             8             8         2           2           2       1
## 2:            5             5             5         8           8           8       2
## 3:          <NA>             8             8      <NA>           3           3       3
## 4:            2             2             2         9           7           7       4
## 5:            7             7             7         2           4           4       5
## 6:            2             2             2         2           2           2       6
```

```r
#### 1) Select only ISCO, as character
isco_data <- data_basis %>%
  dplyr::select(item_id, isco_human1, isco_human2, isco_tool) %>%
  mutate(across(c(isco_human1, isco_human2, isco_tool), as.character))

### Remove all lines with NAs
isco_data <- isco_data %>%
  filter(!is.na(isco_human1) & !is.na(isco_human2) & !is.na(isco_tool))

# First:
### Keep only two most common classes: "2" and "3"

isco_data <- isco_data %>%
  filter(isco_human1 == "2" | isco_human1 == "3") %>%
  filter(isco_human2 == "2" | isco_human2 == "3") %>%
  filter(isco_tool   == "2" | isco_tool   == "3")

# 1) Number of rows where all three classify as "2"
count_all_2 <- isco_data %>%
  dplyr::filter(isco_human1 == "2", isco_human2 == "2", isco_tool == "2") %>%
  nrow()

# 2) Number of rows where all three classify as "3"
count_all_3 <- isco_data %>%
  dplyr::filter(isco_human1 == "3", isco_human2 == "3", isco_tool == "3") %>%
```

```
  nrow()

# 3) Number of rows where at least one value is "2" and at least one value is "3"
count_mixed_2_and_3 <- isco_data %>%
  dplyr::rowwise() %>%
  dplyr::filter(
    any(c(isco_human1, isco_human2, isco_tool) == "2") &
      any(c(isco_human1, isco_human2, isco_tool) == "3")
  ) %>%
  nrow()

# 4) Number of rows where minimum one element is neither "2" nor "3" - should be 0
count_other <- isco_data %>%
  dplyr::rowwise() %>%
  dplyr::filter(
    any(!c(isco_human1, isco_human2, isco_tool) %in% c("2", "3"))
  ) %>%
  nrow()

# Show results
count_all_2         # 235
```

```
## [1] 235
```

```
count_all_3         # 80
```

```
## [1] 80
```

```
count_mixed_2_and_3 # 42
```

```
## [1] 42
```

```
count_other         # 0
```

```
## [1] 0
```

```
# Reindex item_id (1, 2, 3,...)
isco_data <- isco_data %>%
  dplyr::mutate(item_id = dplyr::row_number())

# Collect all observed classes
isco_levels <- isco_data %>%
  dplyr::select(isco_human1, isco_human2, isco_tool) %>%
  unlist(use.names = FALSE) %>%
  unique() %>%
  sort() %>%
  as.character() %>%
  as.vector()

#head(isco_data)

# K, number of distinct classes
K <- length(isco_levels)
stopifnot(K >= 2)

# Helper: map original labels to 1..K using isco_levels for stan compatibility
to_index <- function(v) as.integer(factor(v, levels = isco_levels))
```

```r
# 1) Map original labels to 1..K
isco_data_idx <- isco_data %>%
  dplyr::mutate(
    isco_human1 = to_index(isco_human1),
    isco_human2 = to_index(isco_human2),
    isco_tool   = to_index(isco_tool)
  )

# 2) Long format (now without NAs)
long_data <- isco_data_idx %>%
  dplyr::transmute(item_id, r1 = isco_human1, r2 = isco_human2, r3 = isco_tool) %>%
  tidyr::pivot_longer(cols = starts_with("r"), names_to = "rater", values_to = "y") %>%
  dplyr::mutate(rater_id = as.integer(factor(rater, levels = c("r1", "r2", "r3")))) %>%
  dplyr::arrange(item_id, rater_id)

# Robust 1..I indexing
item_ids <- sort(unique(long_data$item_id))
long_data$item_id_idx <- as.integer(factor(long_data$item_id, levels = item_ids))

I <- length(item_ids) # number of items (people interviewed)
J <- 3L # number of raters
N <- nrow(long_data) # number of observations (I * J if no missing)

cat("Items after filtering:", I, "\n") #357
```

```
## Items after filtering: 357
```

```r
cat("Observations:", N, "\n") #3*357=1071
```

```
## Observations: 1071
```

```r
cat("Classes:", K, "\n") #2
```

```
## Classes: 2
```

```r
# Packages
library(data.table)
library(dplyr)
library(tidyr)
library(cmdstanr)



# Fully neutral Stan data
stan_data <- list(
  I = I,
  J = J,
  K = K,
  N = N,
  ii = as.integer(long_data$item_id_idx),
  jj = as.integer(long_data$rater_id),
  y  = as.integer(long_data$y),

  # No data-driven information
  alpha = rep(1.0, K),
```

```
  beta  = matrix(1.0, nrow = K, ncol = K)
)

# 3) Write Stan model to file
stan_file <- "dawid_skene_isco.stan"
writeLines(con = stan_file, text = '
  data {
    int<lower=2> K;                       // number of categories
    int<lower=1> I;                       // items
    int<lower=1> J;                       // raters
    int<lower=1> N;                       // number of observed ratings
    array[N] int<lower=1, upper=I> ii;    // item index per observation
    array[N] int<lower=1, upper=J> jj;    // rater index per observation
    array[N] int<lower=1, upper=K> y;     // observed category per observation

    vector<lower=0>[K] alpha;             // prior for pi
    array[K] vector<lower=0>[K] beta;     // Dirichlet priors for theta rows, vary by true class
  }

parameters {
  simplex[K] pi;                         // prevalence over true classes
  array[J, K] simplex[K] theta;          // confusion rows: for rater j and true class k
}

transformed parameters {
  // log_q_z[i, k] = log p(z_i = k | pi, theta) up to a constant
  array[I] vector[K] log_q_z;

  // Initialize with log pi
  for (i in 1:I) {
    log_q_z[i] = log(pi);
  }

  // Accumulate per observation
  for (n in 1:N) {
    int i = ii[n];
    int j = jj[n];
    int y_n = y[n];
    for (k in 1:K) {
      // Add log categorical pmf of y_n under theta[j, k, :]
      log_q_z[i, k] += categorical_lpmf(y_n | theta[j, k]);
    }
  }
}

model {
  // Priors
  pi ~ dirichlet(alpha);
  for (j in 1:J)
    for (k in 1:K)
      theta[j, k] ~ dirichlet(beta[k]);

  // Marginalized likelihood
```

```
  for (i in 1:I) {
    target += log_sum_exp(log_q_z[i]);
  }
}

generated quantities {
  array[I] simplex[K] q_z;
  for (i in 1:I) {
    vector[K] lq = log_q_z[i];
    q_z[i] = softmax(lq);
  }
}
')

# 4) Compile and sample with cmdstanr
mod <- cmdstan_model(stan_file, cpp_options = list(stan_threads = TRUE))

fit_basis <- mod$sample(
  data = stan_data,
  seed = 1234,
  chains = 4,
  parallel_chains = 4,
  threads_per_chain = 1,
  iter_warmup = 1000,
  iter_sampling = 4000,
  refresh = 200,
  adapt_delta = 0.99
)
```

```
## Running MCMC with 4 parallel chains, with 1 thread(s) per chain...
##
## Chain 1 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 2 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 3 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 4 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 1 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 2 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 3 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 1 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 4 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 3 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 2 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 4 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 1 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 3 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 2 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 1 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 4 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 3 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 2 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 4 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 1 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 1 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 3 Iteration: 1000 / 5000 [ 20%]  (Warmup)
```

```
## Chain 3 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 2 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 2 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 4 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 1 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 4 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 3 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 1 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 2 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 4 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 1 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 3 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 2 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 4 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 1 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 3 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 2 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 1 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 4 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 2 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 3 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 1 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 4 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 2 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 3 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 1 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 1 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 3 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 2 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 4 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 1 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 2 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 3 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 4 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 1 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 2 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 3 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 1 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 4 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 2 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 3 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 1 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 4 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 2 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 1 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 3 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 4 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 1 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 2 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 3 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 1 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 2 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 3 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 4 Iteration: 3000 / 5000 [ 60%]  (Sampling)
```

```
## Chain 1 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 2 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 3 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 4 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 1 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 2 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 3 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 1 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 4 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 1 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 2 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 3 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 4 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 1 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 1 finished in 43.6 seconds.
## Chain 2 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 3 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 4 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 2 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 3 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 2 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 4 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 3 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 2 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 3 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 4 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 2 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 2 finished in 52.5 seconds.
## Chain 3 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 3 finished in 53.0 seconds.
## Chain 4 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 4 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 4 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 4 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 4 finished in 61.1 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 52.5 seconds.
## Total execution time: 61.1 seconds.
```

```
# print number of rhats above 1.1
s <- fit_basis$summary()
s$rhat <- as.numeric(s$rhat)

subset(s, rhat > 1.1 & grepl("^(pi|theta)\\b", variable))[, c("variable","rhat","ess_bulk","ess_tail")]
```

```
## # A tibble: 0 x 4
## # i 4 variables: variable <chr>, rhat <dbl>, ess_bulk <dbl>, ess_tail <dbl>
```

```
# print string if no rhats above 1.1
if (nrow(subset(s, rhat > 1.1 & grepl("^(pi|theta)\\b", variable))) == 0) {
  cat("No R-hat values above 1.1\n")
}
```

```
## No R-hat values above 1.1
```

```r
fit_chains_lower <- mod$sample(
  data = stan_data,
  seed = 1234,
  chains = 2,
  parallel_chains = 4,
  threads_per_chain = 1,
  iter_warmup = 1000,
  iter_sampling = 4000,
  refresh = 200,
  adapt_delta = 0.99
)
```

```
## Running MCMC with 2 chains, at most 4 in parallel, with 1 thread(s) per chain...
##
## Chain 1 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 2 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 1 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 2 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 1 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 2 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 1 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 2 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 1 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 2 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 1 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 1 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 2 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 2 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 1 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 1 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 2 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 1 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 2 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 1 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 2 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 1 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 2 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 1 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 2 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 1 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 2 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 1 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 1 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 2 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 1 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 2 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 1 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 2 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 1 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 1 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 2 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 1 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 2 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 1 Iteration: 4000 / 5000 [ 80%]  (Sampling)
```

```
## Chain 2 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 1 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 2 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 1 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 2 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 1 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 2 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 1 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 1 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 1 finished in 42.4 seconds.
## Chain 2 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 2 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 2 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 2 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 2 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 2 finished in 50.8 seconds.
##
## Both chains finished successfully.
## Mean chain execution time: 46.6 seconds.
## Total execution time: 50.8 seconds.
```

```
# print number of rhats above 1.1
s <- fit_chains_lower$summary()
s$rhat <- as.numeric(s$rhat)

subset(s, rhat > 1.1 & grepl("^(pi|theta)\\b", variable))[, c("variable","rhat","ess_bulk","ess_tail")]
```

```
## # A tibble: 0 x 4
## # i 4 variables: variable <chr>, rhat <dbl>, ess_bulk <dbl>, ess_tail <dbl>
```

```
# print string if no rhats above 1.1
if (nrow(subset(s, rhat > 1.1 & grepl("^(pi|theta)\\b", variable))) == 0) {
  cat("No R-hat values above 1.1\n")
}
```

```
## No R-hat values above 1.1
```

```
fit_chains_higher <- mod$sample(
  data = stan_data,
  seed = 1234,
  chains = 6,
  parallel_chains = 4,
  threads_per_chain = 1,
  iter_warmup = 1000,
  iter_sampling = 4000,
  refresh = 200,
  adapt_delta = 0.99
)
```

```
## Running MCMC with 6 chains, at most 4 in parallel, with 1 thread(s) per chain...
##
## Chain 1 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 2 Iteration:    1 / 5000 [  0%]  (Warmup)
```

```
## Chain 3 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 4 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 1 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 2 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 3 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 1 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 4 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 3 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 2 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 4 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 1 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 3 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 2 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 1 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 4 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 3 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 2 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 4 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 1 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 1 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 3 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 3 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 2 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 2 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 4 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 1 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 4 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 3 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 1 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 2 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 4 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 1 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 3 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 2 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 4 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 1 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 3 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 2 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 1 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 4 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 2 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 3 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 1 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 4 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 2 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 3 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 1 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 1 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 3 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 2 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 4 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 1 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 2 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 3 Iteration: 2400 / 5000 [ 48%]  (Sampling)
```

```
## Chain 4 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 1 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 2 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 3 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 1 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 4 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 2 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 3 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 1 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 4 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 2 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 3 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 1 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 4 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 1 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 2 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 3 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 1 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 2 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 4 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 3 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 1 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 2 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 3 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 4 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 1 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 2 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 3 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 1 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 4 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 2 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 3 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 1 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 4 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 1 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 1 finished in 43.7 seconds.
## Chain 5 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 2 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 3 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 4 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 2 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 3 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 5 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 2 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 4 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 3 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 5 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 2 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 3 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 5 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 4 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 5 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 2 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 2 finished in 52.7 seconds.
```

```
## Chain 6 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 3 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 3 finished in 53.1 seconds.
## Chain 4 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 5 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 5 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 5 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 6 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 4 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 6 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 5 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 6 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 4 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 5 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 6 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 5 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 4 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 4 finished in 61.5 seconds.
## Chain 6 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 6 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 5 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 6 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 6 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 5 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 6 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 5 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 6 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 5 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 6 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 5 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 6 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 5 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 6 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 6 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 5 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 6 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 5 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 6 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 5 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 6 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 5 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 6 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 5 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 6 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 6 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 5 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 6 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 5 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 6 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 5 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 6 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 5 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 6 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 5 Iteration: 5000 / 5000 [100%]  (Sampling)
```

```
## Chain 5 finished in 47.2 seconds.
## Chain 6 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 6 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 6 finished in 40.0 seconds.
##
## All 6 chains finished successfully.
## Mean chain execution time: 49.7 seconds.
## Total execution time: 92.9 seconds.
```

```
# print number of rhats above 1.1
s <- fit_chains_higher$summary()
s$rhat <- as.numeric(s$rhat)

subset(s, rhat > 1.1 & grepl("^(pi|theta)\\b", variable))[, c("variable","rhat","ess_bulk","ess_tail")]
```

```
## # A tibble: 14 x 4
##    variable      rhat ess_bulk ess_tail
##    <chr>        <dbl>    <dbl>    <dbl>
##  1 pi[1]         1.61     9.84     64.6
##  2 pi[2]         1.61     9.84     64.6
##  3 theta[1,1,1]  1.61     9.85     65.2
##  4 theta[2,1,1]  1.61     9.86     69.0
##  5 theta[3,1,1]  1.61     9.84     67.3
##  6 theta[1,2,1]  1.61     9.86     64.5
##  7 theta[2,2,1]  1.61     9.86     65.1
##  8 theta[3,2,1]  1.61     9.85     66.0
##  9 theta[1,1,2]  1.61     9.85     65.2
## 10 theta[2,1,2]  1.61     9.86     69.0
## 11 theta[3,1,2]  1.61     9.84     67.3
## 12 theta[1,2,2]  1.61     9.86     64.5
## 13 theta[2,2,2]  1.61     9.86     65.1
## 14 theta[3,2,2]  1.61     9.85     66.0
```

```
# print string if no rhats above 1.1
if (nrow(subset(s, rhat > 1.1 & grepl("^(pi|theta)\\b", variable))) == 0) {
  cat("No R-hat values above 1.1\n")
}
```

```
fit_paral_chains_lower <- mod$sample(
  data = stan_data,
  seed = 1234,
  chains = 4,
  parallel_chains = 2,
  threads_per_chain = 1,
  iter_warmup = 1000,
  iter_sampling = 4000,
  refresh = 200,
  adapt_delta = 0.99
)
```

```
## Running MCMC with 4 chains, at most 2 in parallel, with 1 thread(s) per chain...
##
```

```
## Chain 1 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 2 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 1 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 2 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 1 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 2 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 1 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 2 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 1 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 2 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 1 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 1 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 2 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 2 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 1 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 1 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 2 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 1 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 2 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 1 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 2 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 1 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 2 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 1 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 2 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 1 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 1 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 2 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 1 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 2 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 1 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 2 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 1 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 2 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 1 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 2 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 1 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 1 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 2 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 1 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 2 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 1 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 2 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 1 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 2 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 1 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 1 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 2 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 1 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 1 finished in 42.2 seconds.
## Chain 3 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 2 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 2 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 3 Iteration:  200 / 5000 [  4%]  (Warmup)
```

```
## Chain 3 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 2 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 3 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 2 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 3 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 2 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 2 finished in 50.8 seconds.
## Chain 4 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 3 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 3 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 3 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 4 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 3 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 4 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 4 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 3 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 4 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 3 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 4 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 4 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 3 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 4 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 3 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 4 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 3 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 3 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 4 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 3 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 4 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 3 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 4 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 3 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 4 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 3 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 4 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 3 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 3 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 4 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 3 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 4 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 3 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 4 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 3 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 4 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 3 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 4 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 3 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 3 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 3 finished in 51.2 seconds.
## Chain 4 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 4 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 4 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 4 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 4 Iteration: 4400 / 5000 [ 88%]  (Sampling)
```

```
## Chain 4 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 4 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 4 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 4 finished in 60.4 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 51.2 seconds.
## Total execution time: 111.4 seconds.
```

```
# print number of rhats above 1.1
s <- fit_paral_chains_lower$summary()
s$rhat <- as.numeric(s$rhat)

subset(s, rhat > 1.1 & grepl("^(pi|theta)\\b", variable))[, c("variable","rhat","ess_bulk","ess_tail")]
```

```
## # A tibble: 0 x 4
## # i 4 variables: variable <chr>, rhat <dbl>, ess_bulk <dbl>, ess_tail <dbl>
```

```
# print string if no rhats above 1.1
if (nrow(subset(s, rhat > 1.1 & grepl("^(pi|theta)\\b", variable))) == 0) {
  cat("No R-hat values above 1.1\n")
}
```

```
## No R-hat values above 1.1
```

```
fit_paral_chains_higher <- mod$sample(
  data = stan_data,
  seed = 1234,
  chains = 4,
  parallel_chains = 4,
  threads_per_chain = 1,
  iter_warmup = 1000,
  iter_sampling = 4000,
  refresh = 200,
  adapt_delta = 0.99
)
```

```
## Running MCMC with 4 parallel chains, with 1 thread(s) per chain...
##
## Chain 1 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 2 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 3 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 4 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 1 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 2 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 3 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 1 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 4 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 3 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 2 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 4 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 1 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 3 Iteration:  600 / 5000 [ 12%]  (Warmup)
```

```
## Chain 2 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 1 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 4 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 1 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 2 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 3 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 4 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 1 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 3 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 3 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 2 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 2 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 4 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 1 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 4 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 3 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 1 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 2 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 4 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 1 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 3 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 2 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 1 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 4 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 3 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 2 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 1 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 4 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 3 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 2 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 1 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 3 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 2 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 4 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 1 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 3 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 1 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 2 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 4 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 1 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 3 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 2 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 4 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 1 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 3 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 2 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 1 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 4 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 3 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 2 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 1 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 4 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 2 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 3 Iteration: 3000 / 5000 [ 60%]  (Sampling)
```

```
## Chain 1 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 1 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 3 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 2 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 4 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 1 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 2 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 3 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 4 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 1 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 2 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 3 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 1 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 4 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 3 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 2 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 1 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 4 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 3 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 1 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 2 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 1 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 1 finished in 43.5 seconds.
## Chain 4 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 3 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 2 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 3 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 2 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 4 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 2 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 3 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 4 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 2 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 3 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 4 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 2 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 2 finished in 52.5 seconds.
## Chain 3 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 3 finished in 52.6 seconds.
## Chain 4 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 4 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 4 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 4 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 4 finished in 61.9 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 52.6 seconds.
## Total execution time: 62.1 seconds.
```

```r
# print number of rhats above 1.1
s <- fit_paral_chains_higher$summary()
s$rhat <- as.numeric(s$rhat)

subset(s, rhat > 1.1 & grepl("^(pi|theta)\\b", variable))[, c("variable","rhat","ess_bulk","ess_tail")]
```

```
## # A tibble: 0 x 4
## # i 4 variables: variable <chr>, rhat <dbl>, ess_bulk <dbl>, ess_tail <dbl>
```

```
# print string if no rhats above 1.1
if (nrow(subset(s, rhat > 1.1 & grepl("^(pi|theta)\\b", variable))) == 0) {
  cat("No R-hat values above 1.1\n")
}
```

```
## No R-hat values above 1.1
```

```
fit_warmup_lower <- mod$sample(
  data = stan_data,
  seed = 1234,
  chains = 4,
  parallel_chains = 4,
  threads_per_chain = 1,
  iter_warmup = 500,
  iter_sampling = 4000,
  refresh = 200,
  adapt_delta = 0.99
)
```

```
## Running MCMC with 4 parallel chains, with 1 thread(s) per chain...
##
## Chain 1 Iteration:    1 / 4500 [  0%]  (Warmup)
## Chain 1 Iteration:  200 / 4500 [  4%]  (Warmup)
## Chain 2 Iteration:    1 / 4500 [  0%]  (Warmup)
## Chain 2 Iteration:  200 / 4500 [  4%]  (Warmup)
## Chain 3 Iteration:    1 / 4500 [  0%]  (Warmup)
## Chain 4 Iteration:    1 / 4500 [  0%]  (Warmup)
## Chain 1 Iteration:  400 / 4500 [  8%]  (Warmup)
## Chain 1 Iteration:  501 / 4500 [ 11%]  (Sampling)
## Chain 2 Iteration:  400 / 4500 [  8%]  (Warmup)
## Chain 2 Iteration:  501 / 4500 [ 11%]  (Sampling)
## Chain 3 Iteration:  200 / 4500 [  4%]  (Warmup)
## Chain 4 Iteration:  200 / 4500 [  4%]  (Warmup)
## Chain 1 Iteration:  700 / 4500 [ 15%]  (Sampling)
## Chain 3 Iteration:  400 / 4500 [  8%]  (Warmup)
## Chain 2 Iteration:  700 / 4500 [ 15%]  (Sampling)
## Chain 4 Iteration:  400 / 4500 [  8%]  (Warmup)
## Chain 3 Iteration:  501 / 4500 [ 11%]  (Sampling)
## Chain 4 Iteration:  501 / 4500 [ 11%]  (Sampling)
## Chain 2 Iteration:  900 / 4500 [ 20%]  (Sampling)
## Chain 1 Iteration:  900 / 4500 [ 20%]  (Sampling)
## Chain 3 Iteration:  700 / 4500 [ 15%]  (Sampling)
## Chain 2 Iteration: 1100 / 4500 [ 24%]  (Sampling)
## Chain 4 Iteration:  700 / 4500 [ 15%]  (Sampling)
## Chain 1 Iteration: 1100 / 4500 [ 24%]  (Sampling)
## Chain 2 Iteration: 1300 / 4500 [ 28%]  (Sampling)
## Chain 3 Iteration:  900 / 4500 [ 20%]  (Sampling)
## Chain 4 Iteration:  900 / 4500 [ 20%]  (Sampling)
## Chain 1 Iteration: 1300 / 4500 [ 28%]  (Sampling)
```

```
## Chain 2 Iteration: 1500 / 4500 [ 33%]  (Sampling)
## Chain 3 Iteration: 1100 / 4500 [ 24%]  (Sampling)
## Chain 4 Iteration: 1100 / 4500 [ 24%]  (Sampling)
## Chain 2 Iteration: 1700 / 4500 [ 37%]  (Sampling)
## Chain 1 Iteration: 1500 / 4500 [ 33%]  (Sampling)
## Chain 3 Iteration: 1300 / 4500 [ 28%]  (Sampling)
## Chain 2 Iteration: 1900 / 4500 [ 42%]  (Sampling)
## Chain 4 Iteration: 1300 / 4500 [ 28%]  (Sampling)
## Chain 3 Iteration: 1500 / 4500 [ 33%]  (Sampling)
## Chain 1 Iteration: 1700 / 4500 [ 37%]  (Sampling)
## Chain 2 Iteration: 2100 / 4500 [ 46%]  (Sampling)
## Chain 4 Iteration: 1500 / 4500 [ 33%]  (Sampling)
## Chain 3 Iteration: 1700 / 4500 [ 37%]  (Sampling)
## Chain 2 Iteration: 2300 / 4500 [ 51%]  (Sampling)
## Chain 1 Iteration: 1900 / 4500 [ 42%]  (Sampling)
## Chain 4 Iteration: 1700 / 4500 [ 37%]  (Sampling)
## Chain 2 Iteration: 2500 / 4500 [ 55%]  (Sampling)
## Chain 3 Iteration: 1900 / 4500 [ 42%]  (Sampling)
## Chain 1 Iteration: 2100 / 4500 [ 46%]  (Sampling)
## Chain 2 Iteration: 2700 / 4500 [ 60%]  (Sampling)
## Chain 4 Iteration: 1900 / 4500 [ 42%]  (Sampling)
## Chain 3 Iteration: 2100 / 4500 [ 46%]  (Sampling)
## Chain 1 Iteration: 2300 / 4500 [ 51%]  (Sampling)
## Chain 2 Iteration: 2900 / 4500 [ 64%]  (Sampling)
## Chain 3 Iteration: 2300 / 4500 [ 51%]  (Sampling)
## Chain 4 Iteration: 2100 / 4500 [ 46%]  (Sampling)
## Chain 2 Iteration: 3100 / 4500 [ 68%]  (Sampling)
## Chain 1 Iteration: 2500 / 4500 [ 55%]  (Sampling)
## Chain 3 Iteration: 2500 / 4500 [ 55%]  (Sampling)
## Chain 4 Iteration: 2300 / 4500 [ 51%]  (Sampling)
## Chain 2 Iteration: 3300 / 4500 [ 73%]  (Sampling)
## Chain 1 Iteration: 2700 / 4500 [ 60%]  (Sampling)
## Chain 3 Iteration: 2700 / 4500 [ 60%]  (Sampling)
## Chain 2 Iteration: 3500 / 4500 [ 77%]  (Sampling)
## Chain 4 Iteration: 2500 / 4500 [ 55%]  (Sampling)
## Chain 1 Iteration: 2900 / 4500 [ 64%]  (Sampling)
## Chain 2 Iteration: 3700 / 4500 [ 82%]  (Sampling)
## Chain 3 Iteration: 2900 / 4500 [ 64%]  (Sampling)
## Chain 4 Iteration: 2700 / 4500 [ 60%]  (Sampling)
## Chain 2 Iteration: 3900 / 4500 [ 86%]  (Sampling)
## Chain 3 Iteration: 3100 / 4500 [ 68%]  (Sampling)
## Chain 1 Iteration: 3100 / 4500 [ 68%]  (Sampling)
## Chain 4 Iteration: 2900 / 4500 [ 64%]  (Sampling)
## Chain 2 Iteration: 4100 / 4500 [ 91%]  (Sampling)
## Chain 3 Iteration: 3300 / 4500 [ 73%]  (Sampling)
## Chain 1 Iteration: 3300 / 4500 [ 73%]  (Sampling)
## Chain 2 Iteration: 4300 / 4500 [ 95%]  (Sampling)
## Chain 4 Iteration: 3100 / 4500 [ 68%]  (Sampling)
## Chain 3 Iteration: 3500 / 4500 [ 77%]  (Sampling)
## Chain 2 Iteration: 4500 / 4500 [100%]  (Sampling)
## Chain 1 Iteration: 3500 / 4500 [ 77%]  (Sampling)
## Chain 2 finished in 40.3 seconds.
## Chain 4 Iteration: 3300 / 4500 [ 73%]  (Sampling)
## Chain 3 Iteration: 3700 / 4500 [ 82%]  (Sampling)
```

```
## Chain 1 Iteration: 3700 / 4500 [ 82%]  (Sampling)
## Chain 4 Iteration: 3500 / 4500 [ 77%]  (Sampling)
## Chain 3 Iteration: 3900 / 4500 [ 86%]  (Sampling)
## Chain 1 Iteration: 3900 / 4500 [ 86%]  (Sampling)
## Chain 4 Iteration: 3700 / 4500 [ 82%]  (Sampling)
## Chain 3 Iteration: 4100 / 4500 [ 91%]  (Sampling)
## Chain 1 Iteration: 4100 / 4500 [ 91%]  (Sampling)
## Chain 3 Iteration: 4300 / 4500 [ 95%]  (Sampling)
## Chain 4 Iteration: 3900 / 4500 [ 86%]  (Sampling)
## Chain 3 Iteration: 4500 / 4500 [100%]  (Sampling)
## Chain 4 Iteration: 4100 / 4500 [ 91%]  (Sampling)
## Chain 3 finished in 47.2 seconds.
## Chain 1 Iteration: 4300 / 4500 [ 95%]  (Sampling)
## Chain 4 Iteration: 4300 / 4500 [ 95%]  (Sampling)
## Chain 1 Iteration: 4500 / 4500 [100%]  (Sampling)
## Chain 1 finished in 52.5 seconds.
## Chain 4 Iteration: 4500 / 4500 [100%]  (Sampling)
## Chain 4 finished in 51.7 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 47.9 seconds.
## Total execution time: 54.6 seconds.
```

```
# print number of rhats above 1.1
s <- fit_warmup_lower$summary()
s$rhat <- as.numeric(s$rhat)

subset(s, rhat > 1.1 & grepl("^(pi|theta)\\b", variable))[, c("variable","rhat","ess_bulk","ess_tail")]
```

```
## # A tibble: 0 x 4
## # i 4 variables: variable <chr>, rhat <dbl>, ess_bulk <dbl>, ess_tail <dbl>
```

```
# print string if no rhats above 1.1
if (nrow(subset(s, rhat > 1.1 & grepl("^(pi|theta)\\b", variable))) == 0) {
  cat("No R-hat values above 1.1\n")
}
```

```
## No R-hat values above 1.1
```

```
fit_warmup_higher <- mod$sample(
  data = stan_data,
  seed = 1234,
  chains = 4,
  parallel_chains = 4,
  threads_per_chain = 1,
  iter_warmup = 2000,
  iter_sampling = 4000,
  refresh = 200,
  adapt_delta = 0.99
)
```

```
## Running MCMC with 4 parallel chains, with 1 thread(s) per chain...
##
```

```
## Chain 1 Iteration:    1 / 6000 [  0%]  (Warmup)
## Chain 2 Iteration:    1 / 6000 [  0%]  (Warmup)
## Chain 3 Iteration:    1 / 6000 [  0%]  (Warmup)
## Chain 4 Iteration:    1 / 6000 [  0%]  (Warmup)
## Chain 1 Iteration:  200 / 6000 [  3%]  (Warmup)
## Chain 2 Iteration:  200 / 6000 [  3%]  (Warmup)
## Chain 3 Iteration:  200 / 6000 [  3%]  (Warmup)
## Chain 1 Iteration:  400 / 6000 [  6%]  (Warmup)
## Chain 4 Iteration:  200 / 6000 [  3%]  (Warmup)
## Chain 3 Iteration:  400 / 6000 [  6%]  (Warmup)
## Chain 2 Iteration:  400 / 6000 [  6%]  (Warmup)
## Chain 4 Iteration:  400 / 6000 [  6%]  (Warmup)
## Chain 1 Iteration:  600 / 6000 [ 10%]  (Warmup)
## Chain 3 Iteration:  600 / 6000 [ 10%]  (Warmup)
## Chain 2 Iteration:  600 / 6000 [ 10%]  (Warmup)
## Chain 1 Iteration:  800 / 6000 [ 13%]  (Warmup)
## Chain 4 Iteration:  600 / 6000 [ 10%]  (Warmup)
## Chain 3 Iteration:  800 / 6000 [ 13%]  (Warmup)
## Chain 2 Iteration:  800 / 6000 [ 13%]  (Warmup)
## Chain 1 Iteration: 1000 / 6000 [ 16%]  (Warmup)
## Chain 4 Iteration:  800 / 6000 [ 13%]  (Warmup)
## Chain 3 Iteration: 1000 / 6000 [ 16%]  (Warmup)
## Chain 1 Iteration: 1200 / 6000 [ 20%]  (Warmup)
## Chain 2 Iteration: 1000 / 6000 [ 16%]  (Warmup)
## Chain 4 Iteration: 1000 / 6000 [ 16%]  (Warmup)
## Chain 3 Iteration: 1200 / 6000 [ 20%]  (Warmup)
## Chain 1 Iteration: 1400 / 6000 [ 23%]  (Warmup)
## Chain 2 Iteration: 1200 / 6000 [ 20%]  (Warmup)
## Chain 4 Iteration: 1200 / 6000 [ 20%]  (Warmup)
## Chain 3 Iteration: 1400 / 6000 [ 23%]  (Warmup)
## Chain 1 Iteration: 1600 / 6000 [ 26%]  (Warmup)
## Chain 2 Iteration: 1400 / 6000 [ 23%]  (Warmup)
## Chain 4 Iteration: 1400 / 6000 [ 23%]  (Warmup)
## Chain 3 Iteration: 1600 / 6000 [ 26%]  (Warmup)
## Chain 1 Iteration: 1800 / 6000 [ 30%]  (Warmup)
## Chain 2 Iteration: 1600 / 6000 [ 26%]  (Warmup)
## Chain 4 Iteration: 1600 / 6000 [ 26%]  (Warmup)
## Chain 3 Iteration: 1800 / 6000 [ 30%]  (Warmup)
## Chain 1 Iteration: 2000 / 6000 [ 33%]  (Warmup)
## Chain 1 Iteration: 2001 / 6000 [ 33%]  (Sampling)
## Chain 2 Iteration: 1800 / 6000 [ 30%]  (Warmup)
## Chain 4 Iteration: 1800 / 6000 [ 30%]  (Warmup)
## Chain 1 Iteration: 2200 / 6000 [ 36%]  (Sampling)
## Chain 3 Iteration: 2000 / 6000 [ 33%]  (Warmup)
## Chain 4 Iteration: 2000 / 6000 [ 33%]  (Warmup)
## Chain 3 Iteration: 2001 / 6000 [ 33%]  (Sampling)
## Chain 4 Iteration: 2001 / 6000 [ 33%]  (Sampling)
## Chain 2 Iteration: 2000 / 6000 [ 33%]  (Warmup)
## Chain 2 Iteration: 2001 / 6000 [ 33%]  (Sampling)
## Chain 1 Iteration: 2400 / 6000 [ 40%]  (Sampling)
## Chain 4 Iteration: 2200 / 6000 [ 36%]  (Sampling)
## Chain 3 Iteration: 2200 / 6000 [ 36%]  (Sampling)
## Chain 2 Iteration: 2200 / 6000 [ 36%]  (Sampling)
## Chain 1 Iteration: 2600 / 6000 [ 43%]  (Sampling)
```

```
## Chain 4 Iteration: 2400 / 6000 [ 40%]  (Sampling)
## Chain 3 Iteration: 2400 / 6000 [ 40%]  (Sampling)
## Chain 1 Iteration: 2800 / 6000 [ 46%]  (Sampling)
## Chain 2 Iteration: 2400 / 6000 [ 40%]  (Sampling)
## Chain 4 Iteration: 2600 / 6000 [ 43%]  (Sampling)
## Chain 1 Iteration: 3000 / 6000 [ 50%]  (Sampling)
## Chain 4 Iteration: 2800 / 6000 [ 46%]  (Sampling)
## Chain 2 Iteration: 2600 / 6000 [ 43%]  (Sampling)
## Chain 3 Iteration: 2600 / 6000 [ 43%]  (Sampling)
## Chain 1 Iteration: 3200 / 6000 [ 53%]  (Sampling)
## Chain 4 Iteration: 3000 / 6000 [ 50%]  (Sampling)
## Chain 2 Iteration: 2800 / 6000 [ 46%]  (Sampling)
## Chain 3 Iteration: 2800 / 6000 [ 46%]  (Sampling)
## Chain 1 Iteration: 3400 / 6000 [ 56%]  (Sampling)
## Chain 4 Iteration: 3200 / 6000 [ 53%]  (Sampling)
## Chain 1 Iteration: 3600 / 6000 [ 60%]  (Sampling)
## Chain 2 Iteration: 3000 / 6000 [ 50%]  (Sampling)
## Chain 3 Iteration: 3000 / 6000 [ 50%]  (Sampling)
## Chain 4 Iteration: 3400 / 6000 [ 56%]  (Sampling)
## Chain 1 Iteration: 3800 / 6000 [ 63%]  (Sampling)
## Chain 4 Iteration: 3600 / 6000 [ 60%]  (Sampling)
## Chain 2 Iteration: 3200 / 6000 [ 53%]  (Sampling)
## Chain 3 Iteration: 3200 / 6000 [ 53%]  (Sampling)
## Chain 1 Iteration: 4000 / 6000 [ 66%]  (Sampling)
## Chain 4 Iteration: 3800 / 6000 [ 63%]  (Sampling)
## Chain 2 Iteration: 3400 / 6000 [ 56%]  (Sampling)
## Chain 1 Iteration: 4200 / 6000 [ 70%]  (Sampling)
## Chain 3 Iteration: 3400 / 6000 [ 56%]  (Sampling)
## Chain 4 Iteration: 4000 / 6000 [ 66%]  (Sampling)
## Chain 1 Iteration: 4400 / 6000 [ 73%]  (Sampling)
## Chain 2 Iteration: 3600 / 6000 [ 60%]  (Sampling)
## Chain 4 Iteration: 4200 / 6000 [ 70%]  (Sampling)
## Chain 3 Iteration: 3600 / 6000 [ 60%]  (Sampling)
## Chain 1 Iteration: 4600 / 6000 [ 76%]  (Sampling)
## Chain 4 Iteration: 4400 / 6000 [ 73%]  (Sampling)
## Chain 2 Iteration: 3800 / 6000 [ 63%]  (Sampling)
## Chain 3 Iteration: 3800 / 6000 [ 63%]  (Sampling)
## Chain 1 Iteration: 4800 / 6000 [ 80%]  (Sampling)
## Chain 4 Iteration: 4600 / 6000 [ 76%]  (Sampling)
## Chain 1 Iteration: 5000 / 6000 [ 83%]  (Sampling)
## Chain 2 Iteration: 4000 / 6000 [ 66%]  (Sampling)
## Chain 3 Iteration: 4000 / 6000 [ 66%]  (Sampling)
## Chain 4 Iteration: 4800 / 6000 [ 80%]  (Sampling)
## Chain 1 Iteration: 5200 / 6000 [ 86%]  (Sampling)
## Chain 4 Iteration: 5000 / 6000 [ 83%]  (Sampling)
## Chain 2 Iteration: 4200 / 6000 [ 70%]  (Sampling)
## Chain 3 Iteration: 4200 / 6000 [ 70%]  (Sampling)
## Chain 1 Iteration: 5400 / 6000 [ 90%]  (Sampling)
## Chain 4 Iteration: 5200 / 6000 [ 86%]  (Sampling)
## Chain 2 Iteration: 4400 / 6000 [ 73%]  (Sampling)
## Chain 1 Iteration: 5600 / 6000 [ 93%]  (Sampling)
## Chain 3 Iteration: 4400 / 6000 [ 73%]  (Sampling)
## Chain 4 Iteration: 5400 / 6000 [ 90%]  (Sampling)
## Chain 1 Iteration: 5800 / 6000 [ 96%]  (Sampling)
```

```
## Chain 2 Iteration: 4600 / 6000 [ 76%]  (Sampling)
## Chain 4 Iteration: 5600 / 6000 [ 93%]  (Sampling)
## Chain 3 Iteration: 4600 / 6000 [ 76%]  (Sampling)
## Chain 1 Iteration: 6000 / 6000 [100%]  (Sampling)
## Chain 1 finished in 43.5 seconds.
## Chain 4 Iteration: 5800 / 6000 [ 96%]  (Sampling)
## Chain 2 Iteration: 4800 / 6000 [ 80%]  (Sampling)
## Chain 3 Iteration: 4800 / 6000 [ 80%]  (Sampling)
## Chain 4 Iteration: 6000 / 6000 [100%]  (Sampling)
## Chain 4 finished in 45.5 seconds.
## Chain 2 Iteration: 5000 / 6000 [ 83%]  (Sampling)
## Chain 3 Iteration: 5000 / 6000 [ 83%]  (Sampling)
## Chain 2 Iteration: 5200 / 6000 [ 86%]  (Sampling)
## Chain 3 Iteration: 5200 / 6000 [ 86%]  (Sampling)
## Chain 2 Iteration: 5400 / 6000 [ 90%]  (Sampling)
## Chain 3 Iteration: 5400 / 6000 [ 90%]  (Sampling)
## Chain 2 Iteration: 5600 / 6000 [ 93%]  (Sampling)
## Chain 3 Iteration: 5600 / 6000 [ 93%]  (Sampling)
## Chain 2 Iteration: 5800 / 6000 [ 96%]  (Sampling)
## Chain 3 Iteration: 5800 / 6000 [ 96%]  (Sampling)
## Chain 2 Iteration: 6000 / 6000 [100%]  (Sampling)
## Chain 2 finished in 56.3 seconds.
## Chain 3 Iteration: 6000 / 6000 [100%]  (Sampling)
## Chain 3 finished in 57.1 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 50.6 seconds.
## Total execution time: 57.2 seconds.
```

```
# print number of rhats above 1.1
s <- fit_warmup_higher$summary()
s$rhat <- as.numeric(s$rhat)

subset(s, rhat > 1.1 & grepl("^(pi|theta)\\b", variable))[, c("variable","rhat","ess_bulk","ess_tail")]
```

```
## # A tibble: 0 x 4
## # i 4 variables: variable <chr>, rhat <dbl>, ess_bulk <dbl>, ess_tail <dbl>
```

```
# print string if no rhats above 1.1
if (nrow(subset(s, rhat > 1.1 & grepl("^(pi|theta)\\b", variable))) == 0) {
  cat("No R-hat values above 1.1\n")
}
```

```
## No R-hat values above 1.1
```

```
fit_sampling_lower <- mod$sample(
  data = stan_data,
  seed = 1234,
  chains = 4,
  parallel_chains = 4,
  threads_per_chain = 1,
  iter_warmup = 1000,
  iter_sampling = 2000,
```

```
  refresh = 200,
  adapt_delta = 0.99
)
```

```
## Running MCMC with 4 parallel chains, with 1 thread(s) per chain...
##
## Chain 1 Iteration:    1 / 3000 [  0%]  (Warmup)
## Chain 2 Iteration:    1 / 3000 [  0%]  (Warmup)
## Chain 3 Iteration:    1 / 3000 [  0%]  (Warmup)
## Chain 4 Iteration:    1 / 3000 [  0%]  (Warmup)
## Chain 1 Iteration:  200 / 3000 [  6%]  (Warmup)
## Chain 2 Iteration:  200 / 3000 [  6%]  (Warmup)
## Chain 3 Iteration:  200 / 3000 [  6%]  (Warmup)
## Chain 1 Iteration:  400 / 3000 [ 13%]  (Warmup)
## Chain 4 Iteration:  200 / 3000 [  6%]  (Warmup)
## Chain 3 Iteration:  400 / 3000 [ 13%]  (Warmup)
## Chain 1 Iteration:  600 / 3000 [ 20%]  (Warmup)
## Chain 2 Iteration:  400 / 3000 [ 13%]  (Warmup)
## Chain 3 Iteration:  600 / 3000 [ 20%]  (Warmup)
## Chain 4 Iteration:  400 / 3000 [ 13%]  (Warmup)
## Chain 1 Iteration:  800 / 3000 [ 26%]  (Warmup)
## Chain 2 Iteration:  600 / 3000 [ 20%]  (Warmup)
## Chain 4 Iteration:  600 / 3000 [ 20%]  (Warmup)
## Chain 3 Iteration:  800 / 3000 [ 26%]  (Warmup)
## Chain 2 Iteration:  800 / 3000 [ 26%]  (Warmup)
## Chain 4 Iteration:  800 / 3000 [ 26%]  (Warmup)
## Chain 1 Iteration: 1000 / 3000 [ 33%]  (Warmup)
## Chain 1 Iteration: 1001 / 3000 [ 33%]  (Sampling)
## Chain 3 Iteration: 1000 / 3000 [ 33%]  (Warmup)
## Chain 3 Iteration: 1001 / 3000 [ 33%]  (Sampling)
## Chain 2 Iteration: 1000 / 3000 [ 33%]  (Warmup)
## Chain 2 Iteration: 1001 / 3000 [ 33%]  (Sampling)
## Chain 4 Iteration: 1000 / 3000 [ 33%]  (Warmup)
## Chain 4 Iteration: 1001 / 3000 [ 33%]  (Sampling)
## Chain 1 Iteration: 1200 / 3000 [ 40%]  (Sampling)
## Chain 3 Iteration: 1200 / 3000 [ 40%]  (Sampling)
## Chain 2 Iteration: 1200 / 3000 [ 40%]  (Sampling)
## Chain 1 Iteration: 1400 / 3000 [ 46%]  (Sampling)
## Chain 4 Iteration: 1200 / 3000 [ 40%]  (Sampling)
## Chain 1 Iteration: 1600 / 3000 [ 53%]  (Sampling)
## Chain 3 Iteration: 1400 / 3000 [ 46%]  (Sampling)
## Chain 2 Iteration: 1400 / 3000 [ 46%]  (Sampling)
## Chain 4 Iteration: 1400 / 3000 [ 46%]  (Sampling)
## Chain 1 Iteration: 1800 / 3000 [ 60%]  (Sampling)
## Chain 2 Iteration: 1600 / 3000 [ 53%]  (Sampling)
## Chain 3 Iteration: 1600 / 3000 [ 53%]  (Sampling)
## Chain 1 Iteration: 2000 / 3000 [ 66%]  (Sampling)
## Chain 4 Iteration: 1600 / 3000 [ 53%]  (Sampling)
## Chain 2 Iteration: 1800 / 3000 [ 60%]  (Sampling)
## Chain 3 Iteration: 1800 / 3000 [ 60%]  (Sampling)
## Chain 1 Iteration: 2200 / 3000 [ 73%]  (Sampling)
## Chain 4 Iteration: 1800 / 3000 [ 60%]  (Sampling)
## Chain 2 Iteration: 2000 / 3000 [ 66%]  (Sampling)
## Chain 3 Iteration: 2000 / 3000 [ 66%]  (Sampling)
```

```
## Chain 1 Iteration: 2400 / 3000 [ 80%]  (Sampling)
## Chain 1 Iteration: 2600 / 3000 [ 86%]  (Sampling)
## Chain 2 Iteration: 2200 / 3000 [ 73%]  (Sampling)
## Chain 3 Iteration: 2200 / 3000 [ 73%]  (Sampling)
## Chain 4 Iteration: 2000 / 3000 [ 66%]  (Sampling)
## Chain 1 Iteration: 2800 / 3000 [ 93%]  (Sampling)
## Chain 2 Iteration: 2400 / 3000 [ 80%]  (Sampling)
## Chain 3 Iteration: 2400 / 3000 [ 80%]  (Sampling)
## Chain 4 Iteration: 2200 / 3000 [ 73%]  (Sampling)
## Chain 1 Iteration: 3000 / 3000 [100%]  (Sampling)
## Chain 1 finished in 25.7 seconds.
## Chain 2 Iteration: 2600 / 3000 [ 86%]  (Sampling)
## Chain 3 Iteration: 2600 / 3000 [ 86%]  (Sampling)
## Chain 4 Iteration: 2400 / 3000 [ 80%]  (Sampling)
## Chain 2 Iteration: 2800 / 3000 [ 93%]  (Sampling)
## Chain 3 Iteration: 2800 / 3000 [ 93%]  (Sampling)
## Chain 4 Iteration: 2600 / 3000 [ 86%]  (Sampling)
## Chain 2 Iteration: 3000 / 3000 [100%]  (Sampling)
## Chain 2 finished in 30.8 seconds.
## Chain 3 Iteration: 3000 / 3000 [100%]  (Sampling)
## Chain 3 finished in 31.1 seconds.
## Chain 4 Iteration: 2800 / 3000 [ 93%]  (Sampling)
## Chain 4 Iteration: 3000 / 3000 [100%]  (Sampling)
## Chain 4 finished in 35.1 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 30.7 seconds.
## Total execution time: 35.2 seconds.
```

```
# print number of rhats above 1.1
s <- fit_sampling_lower$summary()
s$rhat <- as.numeric(s$rhat)

subset(s, rhat > 1.1 & grepl("^(pi|theta)\\b", variable))[, c("variable","rhat","ess_bulk","ess_tail")]
```

```
## # A tibble: 0 x 4
## # i 4 variables: variable <chr>, rhat <dbl>, ess_bulk <dbl>, ess_tail <dbl>
```

```
# print string if no rhats above 1.1
if (nrow(subset(s, rhat > 1.1 & grepl("^(pi|theta)\\b", variable))) == 0) {
  cat("No R-hat values above 1.1\n")
}
```

```
## No R-hat values above 1.1
```

```
fit_sampling_higher <- mod$sample(
  data = stan_data,
  seed = 1234,
  chains = 4,
  parallel_chains = 4,
  threads_per_chain = 1,
  iter_warmup = 1000,
  iter_sampling = 8000,
```

```
  refresh = 200,
  adapt_delta = 0.99
)
```

```
## Running MCMC with 4 parallel chains, with 1 thread(s) per chain...
##
## Chain 1 Iteration:    1 / 9000 [  0%]  (Warmup)
## Chain 2 Iteration:    1 / 9000 [  0%]  (Warmup)
## Chain 3 Iteration:    1 / 9000 [  0%]  (Warmup)
## Chain 4 Iteration:    1 / 9000 [  0%]  (Warmup)
## Chain 1 Iteration:  200 / 9000 [  2%]  (Warmup)
## Chain 2 Iteration:  200 / 9000 [  2%]  (Warmup)
## Chain 3 Iteration:  200 / 9000 [  2%]  (Warmup)
## Chain 1 Iteration:  400 / 9000 [  4%]  (Warmup)
## Chain 4 Iteration:  200 / 9000 [  2%]  (Warmup)
## Chain 3 Iteration:  400 / 9000 [  4%]  (Warmup)
## Chain 2 Iteration:  400 / 9000 [  4%]  (Warmup)
## Chain 4 Iteration:  400 / 9000 [  4%]  (Warmup)
## Chain 1 Iteration:  600 / 9000 [  6%]  (Warmup)
## Chain 3 Iteration:  600 / 9000 [  6%]  (Warmup)
## Chain 2 Iteration:  600 / 9000 [  6%]  (Warmup)
## Chain 4 Iteration:  600 / 9000 [  6%]  (Warmup)
## Chain 1 Iteration:  800 / 9000 [  8%]  (Warmup)
## Chain 3 Iteration:  800 / 9000 [  8%]  (Warmup)
## Chain 2 Iteration:  800 / 9000 [  8%]  (Warmup)
## Chain 4 Iteration:  800 / 9000 [  8%]  (Warmup)
## Chain 1 Iteration: 1000 / 9000 [ 11%]  (Warmup)
## Chain 1 Iteration: 1001 / 9000 [ 11%]  (Sampling)
## Chain 3 Iteration: 1000 / 9000 [ 11%]  (Warmup)
## Chain 3 Iteration: 1001 / 9000 [ 11%]  (Sampling)
## Chain 2 Iteration: 1000 / 9000 [ 11%]  (Warmup)
## Chain 2 Iteration: 1001 / 9000 [ 11%]  (Sampling)
## Chain 4 Iteration: 1000 / 9000 [ 11%]  (Warmup)
## Chain 4 Iteration: 1001 / 9000 [ 11%]  (Sampling)
## Chain 1 Iteration: 1200 / 9000 [ 13%]  (Sampling)
## Chain 3 Iteration: 1200 / 9000 [ 13%]  (Sampling)
## Chain 2 Iteration: 1200 / 9000 [ 13%]  (Sampling)
## Chain 4 Iteration: 1200 / 9000 [ 13%]  (Sampling)
## Chain 1 Iteration: 1400 / 9000 [ 15%]  (Sampling)
## Chain 3 Iteration: 1400 / 9000 [ 15%]  (Sampling)
## Chain 2 Iteration: 1400 / 9000 [ 15%]  (Sampling)
## Chain 1 Iteration: 1600 / 9000 [ 17%]  (Sampling)
## Chain 4 Iteration: 1400 / 9000 [ 15%]  (Sampling)
## Chain 3 Iteration: 1600 / 9000 [ 17%]  (Sampling)
## Chain 1 Iteration: 1800 / 9000 [ 20%]  (Sampling)
## Chain 2 Iteration: 1600 / 9000 [ 17%]  (Sampling)
## Chain 4 Iteration: 1600 / 9000 [ 17%]  (Sampling)
## Chain 3 Iteration: 1800 / 9000 [ 20%]  (Sampling)
## Chain 1 Iteration: 2000 / 9000 [ 22%]  (Sampling)
## Chain 2 Iteration: 1800 / 9000 [ 20%]  (Sampling)
## Chain 1 Iteration: 2200 / 9000 [ 24%]  (Sampling)
## Chain 3 Iteration: 2000 / 9000 [ 22%]  (Sampling)
## Chain 4 Iteration: 1800 / 9000 [ 20%]  (Sampling)
## Chain 2 Iteration: 2000 / 9000 [ 22%]  (Sampling)
```

```
## Chain 1 Iteration: 2400 / 9000 [ 26%]  (Sampling)
## Chain 3 Iteration: 2200 / 9000 [ 24%]  (Sampling)
## Chain 2 Iteration: 2200 / 9000 [ 24%]  (Sampling)
## Chain 4 Iteration: 2000 / 9000 [ 22%]  (Sampling)
## Chain 1 Iteration: 2600 / 9000 [ 28%]  (Sampling)
## Chain 3 Iteration: 2400 / 9000 [ 26%]  (Sampling)
## Chain 2 Iteration: 2400 / 9000 [ 26%]  (Sampling)
## Chain 4 Iteration: 2200 / 9000 [ 24%]  (Sampling)
## Chain 1 Iteration: 2800 / 9000 [ 31%]  (Sampling)
## Chain 3 Iteration: 2600 / 9000 [ 28%]  (Sampling)
## Chain 2 Iteration: 2600 / 9000 [ 28%]  (Sampling)
## Chain 1 Iteration: 3000 / 9000 [ 33%]  (Sampling)
## Chain 4 Iteration: 2400 / 9000 [ 26%]  (Sampling)
## Chain 3 Iteration: 2800 / 9000 [ 31%]  (Sampling)
## Chain 2 Iteration: 2800 / 9000 [ 31%]  (Sampling)
## Chain 1 Iteration: 3200 / 9000 [ 35%]  (Sampling)
## Chain 4 Iteration: 2600 / 9000 [ 28%]  (Sampling)
## Chain 3 Iteration: 3000 / 9000 [ 33%]  (Sampling)
## Chain 1 Iteration: 3400 / 9000 [ 37%]  (Sampling)
## Chain 2 Iteration: 3000 / 9000 [ 33%]  (Sampling)
## Chain 4 Iteration: 2800 / 9000 [ 31%]  (Sampling)
## Chain 3 Iteration: 3200 / 9000 [ 35%]  (Sampling)
## Chain 1 Iteration: 3600 / 9000 [ 40%]  (Sampling)
## Chain 2 Iteration: 3200 / 9000 [ 35%]  (Sampling)
## Chain 1 Iteration: 3800 / 9000 [ 42%]  (Sampling)
## Chain 3 Iteration: 3400 / 9000 [ 37%]  (Sampling)
## Chain 4 Iteration: 3000 / 9000 [ 33%]  (Sampling)
## Chain 2 Iteration: 3400 / 9000 [ 37%]  (Sampling)
## Chain 1 Iteration: 4000 / 9000 [ 44%]  (Sampling)
## Chain 3 Iteration: 3600 / 9000 [ 40%]  (Sampling)
## Chain 2 Iteration: 3600 / 9000 [ 40%]  (Sampling)
## Chain 4 Iteration: 3200 / 9000 [ 35%]  (Sampling)
## Chain 1 Iteration: 4200 / 9000 [ 46%]  (Sampling)
## Chain 3 Iteration: 3800 / 9000 [ 42%]  (Sampling)
## Chain 2 Iteration: 3800 / 9000 [ 42%]  (Sampling)
## Chain 4 Iteration: 3400 / 9000 [ 37%]  (Sampling)
## Chain 1 Iteration: 4400 / 9000 [ 48%]  (Sampling)
## Chain 3 Iteration: 4000 / 9000 [ 44%]  (Sampling)
## Chain 2 Iteration: 4000 / 9000 [ 44%]  (Sampling)
## Chain 1 Iteration: 4600 / 9000 [ 51%]  (Sampling)
## Chain 4 Iteration: 3600 / 9000 [ 40%]  (Sampling)
## Chain 3 Iteration: 4200 / 9000 [ 46%]  (Sampling)
## Chain 1 Iteration: 4800 / 9000 [ 53%]  (Sampling)
## Chain 2 Iteration: 4200 / 9000 [ 46%]  (Sampling)
## Chain 3 Iteration: 4400 / 9000 [ 48%]  (Sampling)
## Chain 4 Iteration: 3800 / 9000 [ 42%]  (Sampling)
## Chain 1 Iteration: 5000 / 9000 [ 55%]  (Sampling)
## Chain 2 Iteration: 4400 / 9000 [ 48%]  (Sampling)
## Chain 3 Iteration: 4600 / 9000 [ 51%]  (Sampling)
## Chain 1 Iteration: 5200 / 9000 [ 57%]  (Sampling)
## Chain 4 Iteration: 4000 / 9000 [ 44%]  (Sampling)
## Chain 2 Iteration: 4600 / 9000 [ 51%]  (Sampling)
## Chain 1 Iteration: 5400 / 9000 [ 60%]  (Sampling)
## Chain 3 Iteration: 4800 / 9000 [ 53%]  (Sampling)
```

```
## Chain 2 Iteration: 4800 / 9000 [ 53%]  (Sampling)
## Chain 4 Iteration: 4200 / 9000 [ 46%]  (Sampling)
## Chain 1 Iteration: 5600 / 9000 [ 62%]  (Sampling)
## Chain 3 Iteration: 5000 / 9000 [ 55%]  (Sampling)
## Chain 2 Iteration: 5000 / 9000 [ 55%]  (Sampling)
## Chain 4 Iteration: 4400 / 9000 [ 48%]  (Sampling)
## Chain 1 Iteration: 5800 / 9000 [ 64%]  (Sampling)
## Chain 3 Iteration: 5200 / 9000 [ 57%]  (Sampling)
## Chain 2 Iteration: 5200 / 9000 [ 57%]  (Sampling)
## Chain 1 Iteration: 6000 / 9000 [ 66%]  (Sampling)
## Chain 4 Iteration: 4600 / 9000 [ 51%]  (Sampling)
## Chain 3 Iteration: 5400 / 9000 [ 60%]  (Sampling)
## Chain 2 Iteration: 5400 / 9000 [ 60%]  (Sampling)
## Chain 1 Iteration: 6200 / 9000 [ 68%]  (Sampling)
## Chain 4 Iteration: 4800 / 9000 [ 53%]  (Sampling)
## Chain 3 Iteration: 5600 / 9000 [ 62%]  (Sampling)
## Chain 2 Iteration: 5600 / 9000 [ 62%]  (Sampling)
## Chain 1 Iteration: 6400 / 9000 [ 71%]  (Sampling)
## Chain 3 Iteration: 5800 / 9000 [ 64%]  (Sampling)
## Chain 4 Iteration: 5000 / 9000 [ 55%]  (Sampling)
## Chain 1 Iteration: 6600 / 9000 [ 73%]  (Sampling)
## Chain 2 Iteration: 5800 / 9000 [ 64%]  (Sampling)
## Chain 3 Iteration: 6000 / 9000 [ 66%]  (Sampling)
## Chain 1 Iteration: 6800 / 9000 [ 75%]  (Sampling)
## Chain 2 Iteration: 6000 / 9000 [ 66%]  (Sampling)
## Chain 4 Iteration: 5200 / 9000 [ 57%]  (Sampling)
## Chain 3 Iteration: 6200 / 9000 [ 68%]  (Sampling)
## Chain 1 Iteration: 7000 / 9000 [ 77%]  (Sampling)
## Chain 2 Iteration: 6200 / 9000 [ 68%]  (Sampling)
## Chain 4 Iteration: 5400 / 9000 [ 60%]  (Sampling)
## Chain 1 Iteration: 7200 / 9000 [ 80%]  (Sampling)
## Chain 3 Iteration: 6400 / 9000 [ 71%]  (Sampling)
## Chain 2 Iteration: 6400 / 9000 [ 71%]  (Sampling)
## Chain 4 Iteration: 5600 / 9000 [ 62%]  (Sampling)
## Chain 1 Iteration: 7400 / 9000 [ 82%]  (Sampling)
## Chain 3 Iteration: 6600 / 9000 [ 73%]  (Sampling)
## Chain 2 Iteration: 6600 / 9000 [ 73%]  (Sampling)
## Chain 1 Iteration: 7600 / 9000 [ 84%]  (Sampling)
## Chain 4 Iteration: 5800 / 9000 [ 64%]  (Sampling)
## Chain 3 Iteration: 6800 / 9000 [ 75%]  (Sampling)
## Chain 2 Iteration: 6800 / 9000 [ 75%]  (Sampling)
## Chain 1 Iteration: 7800 / 9000 [ 86%]  (Sampling)
## Chain 3 Iteration: 7000 / 9000 [ 77%]  (Sampling)
## Chain 4 Iteration: 6000 / 9000 [ 66%]  (Sampling)
## Chain 2 Iteration: 7000 / 9000 [ 77%]  (Sampling)
## Chain 1 Iteration: 8000 / 9000 [ 88%]  (Sampling)
## Chain 3 Iteration: 7200 / 9000 [ 80%]  (Sampling)
## Chain 1 Iteration: 8200 / 9000 [ 91%]  (Sampling)
## Chain 2 Iteration: 7200 / 9000 [ 80%]  (Sampling)
## Chain 4 Iteration: 6200 / 9000 [ 68%]  (Sampling)
## Chain 3 Iteration: 7400 / 9000 [ 82%]  (Sampling)
## Chain 1 Iteration: 8400 / 9000 [ 93%]  (Sampling)
## Chain 2 Iteration: 7400 / 9000 [ 82%]  (Sampling)
## Chain 4 Iteration: 6400 / 9000 [ 71%]  (Sampling)
```

```
## Chain 3 Iteration: 7600 / 9000 [ 84%]  (Sampling)
## Chain 1 Iteration: 8600 / 9000 [ 95%]  (Sampling)
## Chain 2 Iteration: 7600 / 9000 [ 84%]  (Sampling)
## Chain 4 Iteration: 6600 / 9000 [ 73%]  (Sampling)
## Chain 3 Iteration: 7800 / 9000 [ 86%]  (Sampling)
## Chain 1 Iteration: 8800 / 9000 [ 97%]  (Sampling)
## Chain 2 Iteration: 7800 / 9000 [ 86%]  (Sampling)
## Chain 1 Iteration: 9000 / 9000 [100%]  (Sampling)
## Chain 4 Iteration: 6800 / 9000 [ 75%]  (Sampling)
## Chain 1 finished in 85.5 seconds.
## Chain 3 Iteration: 8000 / 9000 [ 88%]  (Sampling)
## Chain 2 Iteration: 8000 / 9000 [ 88%]  (Sampling)
## Chain 3 Iteration: 8200 / 9000 [ 91%]  (Sampling)
## Chain 4 Iteration: 7000 / 9000 [ 77%]  (Sampling)
## Chain 2 Iteration: 8200 / 9000 [ 91%]  (Sampling)
## Chain 3 Iteration: 8400 / 9000 [ 93%]  (Sampling)
## Chain 4 Iteration: 7200 / 9000 [ 80%]  (Sampling)
## Chain 2 Iteration: 8400 / 9000 [ 93%]  (Sampling)
## Chain 3 Iteration: 8600 / 9000 [ 95%]  (Sampling)
## Chain 2 Iteration: 8600 / 9000 [ 95%]  (Sampling)
## Chain 4 Iteration: 7400 / 9000 [ 82%]  (Sampling)
## Chain 3 Iteration: 8800 / 9000 [ 97%]  (Sampling)
## Chain 2 Iteration: 8800 / 9000 [ 97%]  (Sampling)
## Chain 4 Iteration: 7600 / 9000 [ 84%]  (Sampling)
## Chain 3 Iteration: 9000 / 9000 [100%]  (Sampling)
## Chain 3 finished in 96.2 seconds.
## Chain 2 Iteration: 9000 / 9000 [100%]  (Sampling)
## Chain 2 finished in 97.8 seconds.
## Chain 4 Iteration: 7800 / 9000 [ 86%]  (Sampling)
## Chain 4 Iteration: 8000 / 9000 [ 88%]  (Sampling)
## Chain 4 Iteration: 8200 / 9000 [ 91%]  (Sampling)
## Chain 4 Iteration: 8400 / 9000 [ 93%]  (Sampling)
## Chain 4 Iteration: 8600 / 9000 [ 95%]  (Sampling)
## Chain 4 Iteration: 8800 / 9000 [ 97%]  (Sampling)
## Chain 4 Iteration: 9000 / 9000 [100%]  (Sampling)
## Chain 4 finished in 113.4 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 98.2 seconds.
## Total execution time: 113.6 seconds.
```

```
# print number of rhats above 1.1
s <- fit_sampling_higher$summary()
s$rhat <- as.numeric(s$rhat)

subset(s, rhat > 1.1 & grepl("^(pi|theta)\\b", variable))[, c("variable","rhat","ess_bulk","ess_tail")]
```

```
## # A tibble: 0 x 4
## # i 4 variables: variable <chr>, rhat <dbl>, ess_bulk <dbl>, ess_tail <dbl>
```

```
# print string if no rhats above 1.1
if (nrow(subset(s, rhat > 1.1 & grepl("^(pi|theta)\\b", variable))) == 0) {
  cat("No R-hat values above 1.1\n")
}
```

```
## No R-hat values above 1.1
```

```r
fit_refresh_lower <- mod$sample(
  data = stan_data,
  seed = 1234,
  chains = 4,
  parallel_chains = 4,
  threads_per_chain = 1,
  iter_warmup = 1000,
  iter_sampling = 4000,
  refresh = 100,
  adapt_delta = 0.99
)
```

```
## Running MCMC with 4 parallel chains, with 1 thread(s) per chain...
##
## Chain 1 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 2 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 3 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 4 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 1 Iteration:  100 / 5000 [  2%]  (Warmup)
## Chain 3 Iteration:  100 / 5000 [  2%]  (Warmup)
## Chain 2 Iteration:  100 / 5000 [  2%]  (Warmup)
## Chain 4 Iteration:  100 / 5000 [  2%]  (Warmup)
## Chain 1 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 1 Iteration:  300 / 5000 [  6%]  (Warmup)
## Chain 2 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 3 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 1 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 4 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 3 Iteration:  300 / 5000 [  6%]  (Warmup)
## Chain 2 Iteration:  300 / 5000 [  6%]  (Warmup)
## Chain 4 Iteration:  300 / 5000 [  6%]  (Warmup)
## Chain 1 Iteration:  500 / 5000 [ 10%]  (Warmup)
## Chain 3 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 2 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 4 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 1 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 3 Iteration:  500 / 5000 [ 10%]  (Warmup)
## Chain 2 Iteration:  500 / 5000 [ 10%]  (Warmup)
## Chain 1 Iteration:  700 / 5000 [ 14%]  (Warmup)
## Chain 4 Iteration:  500 / 5000 [ 10%]  (Warmup)
## Chain 3 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 2 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 1 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 4 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 3 Iteration:  700 / 5000 [ 14%]  (Warmup)
## Chain 2 Iteration:  700 / 5000 [ 14%]  (Warmup)
## Chain 1 Iteration:  900 / 5000 [ 18%]  (Warmup)
## Chain 4 Iteration:  700 / 5000 [ 14%]  (Warmup)
## Chain 3 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 2 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 4 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 3 Iteration:  900 / 5000 [ 18%]  (Warmup)
```

```
## Chain 1 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 1 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 2 Iteration:  900 / 5000 [ 18%]  (Warmup)
## Chain 4 Iteration:  900 / 5000 [ 18%]  (Warmup)
## Chain 1 Iteration: 1100 / 5000 [ 22%]  (Sampling)
## Chain 3 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 3 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 2 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 2 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 4 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 1 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 4 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 3 Iteration: 1100 / 5000 [ 22%]  (Sampling)
## Chain 2 Iteration: 1100 / 5000 [ 22%]  (Sampling)
## Chain 1 Iteration: 1300 / 5000 [ 26%]  (Sampling)
## Chain 4 Iteration: 1100 / 5000 [ 22%]  (Sampling)
## Chain 3 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 1 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 2 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 3 Iteration: 1300 / 5000 [ 26%]  (Sampling)
## Chain 4 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 1 Iteration: 1500 / 5000 [ 30%]  (Sampling)
## Chain 2 Iteration: 1300 / 5000 [ 26%]  (Sampling)
## Chain 1 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 3 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 4 Iteration: 1300 / 5000 [ 26%]  (Sampling)
## Chain 2 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 1 Iteration: 1700 / 5000 [ 34%]  (Sampling)
## Chain 3 Iteration: 1500 / 5000 [ 30%]  (Sampling)
## Chain 2 Iteration: 1500 / 5000 [ 30%]  (Sampling)
## Chain 4 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 1 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 2 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 3 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 1 Iteration: 1900 / 5000 [ 38%]  (Sampling)
## Chain 4 Iteration: 1500 / 5000 [ 30%]  (Sampling)
## Chain 2 Iteration: 1700 / 5000 [ 34%]  (Sampling)
## Chain 3 Iteration: 1700 / 5000 [ 34%]  (Sampling)
## Chain 1 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 4 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 1 Iteration: 2100 / 5000 [ 42%]  (Sampling)
## Chain 2 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 3 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 4 Iteration: 1700 / 5000 [ 34%]  (Sampling)
## Chain 1 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 2 Iteration: 1900 / 5000 [ 38%]  (Sampling)
## Chain 3 Iteration: 1900 / 5000 [ 38%]  (Sampling)
## Chain 1 Iteration: 2300 / 5000 [ 46%]  (Sampling)
## Chain 4 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 2 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 3 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 1 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 2 Iteration: 2100 / 5000 [ 42%]  (Sampling)
## Chain 3 Iteration: 2100 / 5000 [ 42%]  (Sampling)
```

```
## Chain 4 Iteration: 1900 / 5000 [ 38%]  (Sampling)
## Chain 1 Iteration: 2500 / 5000 [ 50%]  (Sampling)
## Chain 2 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 3 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 1 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 4 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 1 Iteration: 2700 / 5000 [ 54%]  (Sampling)
## Chain 2 Iteration: 2300 / 5000 [ 46%]  (Sampling)
## Chain 3 Iteration: 2300 / 5000 [ 46%]  (Sampling)
## Chain 4 Iteration: 2100 / 5000 [ 42%]  (Sampling)
## Chain 1 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 2 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 3 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 1 Iteration: 2900 / 5000 [ 58%]  (Sampling)
## Chain 4 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 2 Iteration: 2500 / 5000 [ 50%]  (Sampling)
## Chain 3 Iteration: 2500 / 5000 [ 50%]  (Sampling)
## Chain 1 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 4 Iteration: 2300 / 5000 [ 46%]  (Sampling)
## Chain 2 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 3 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 1 Iteration: 3100 / 5000 [ 62%]  (Sampling)
## Chain 4 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 1 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 2 Iteration: 2700 / 5000 [ 54%]  (Sampling)
## Chain 3 Iteration: 2700 / 5000 [ 54%]  (Sampling)
## Chain 1 Iteration: 3300 / 5000 [ 66%]  (Sampling)
## Chain 2 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 3 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 4 Iteration: 2500 / 5000 [ 50%]  (Sampling)
## Chain 1 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 2 Iteration: 2900 / 5000 [ 58%]  (Sampling)
## Chain 3 Iteration: 2900 / 5000 [ 58%]  (Sampling)
## Chain 4 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 1 Iteration: 3500 / 5000 [ 70%]  (Sampling)
## Chain 2 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 3 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 1 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 4 Iteration: 2700 / 5000 [ 54%]  (Sampling)
## Chain 2 Iteration: 3100 / 5000 [ 62%]  (Sampling)
## Chain 1 Iteration: 3700 / 5000 [ 74%]  (Sampling)
## Chain 3 Iteration: 3100 / 5000 [ 62%]  (Sampling)
## Chain 4 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 1 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 2 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 3 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 1 Iteration: 3900 / 5000 [ 78%]  (Sampling)
## Chain 2 Iteration: 3300 / 5000 [ 66%]  (Sampling)
## Chain 4 Iteration: 2900 / 5000 [ 58%]  (Sampling)
## Chain 3 Iteration: 3300 / 5000 [ 66%]  (Sampling)
## Chain 1 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 2 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 4 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 3 Iteration: 3400 / 5000 [ 68%]  (Sampling)
```

```
## Chain 1 Iteration: 4100 / 5000 [ 82%]  (Sampling)
## Chain 2 Iteration: 3500 / 5000 [ 70%]  (Sampling)
## Chain 3 Iteration: 3500 / 5000 [ 70%]  (Sampling)
## Chain 1 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 4 Iteration: 3100 / 5000 [ 62%]  (Sampling)
## Chain 2 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 1 Iteration: 4300 / 5000 [ 86%]  (Sampling)
## Chain 3 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 4 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 2 Iteration: 3700 / 5000 [ 74%]  (Sampling)
## Chain 1 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 3 Iteration: 3700 / 5000 [ 74%]  (Sampling)
## Chain 4 Iteration: 3300 / 5000 [ 66%]  (Sampling)
## Chain 1 Iteration: 4500 / 5000 [ 90%]  (Sampling)
## Chain 2 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 3 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 1 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 4 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 2 Iteration: 3900 / 5000 [ 78%]  (Sampling)
## Chain 3 Iteration: 3900 / 5000 [ 78%]  (Sampling)
## Chain 1 Iteration: 4700 / 5000 [ 94%]  (Sampling)
## Chain 2 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 4 Iteration: 3500 / 5000 [ 70%]  (Sampling)
## Chain 3 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 1 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 2 Iteration: 4100 / 5000 [ 82%]  (Sampling)
## Chain 1 Iteration: 4900 / 5000 [ 98%]  (Sampling)
## Chain 3 Iteration: 4100 / 5000 [ 82%]  (Sampling)
## Chain 4 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 2 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 1 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 1 finished in 43.9 seconds.
## Chain 3 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 4 Iteration: 3700 / 5000 [ 74%]  (Sampling)
## Chain 2 Iteration: 4300 / 5000 [ 86%]  (Sampling)
## Chain 3 Iteration: 4300 / 5000 [ 86%]  (Sampling)
## Chain 4 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 2 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 3 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 2 Iteration: 4500 / 5000 [ 90%]  (Sampling)
## Chain 4 Iteration: 3900 / 5000 [ 78%]  (Sampling)
## Chain 3 Iteration: 4500 / 5000 [ 90%]  (Sampling)
## Chain 2 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 4 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 3 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 2 Iteration: 4700 / 5000 [ 94%]  (Sampling)
## Chain 3 Iteration: 4700 / 5000 [ 94%]  (Sampling)
## Chain 4 Iteration: 4100 / 5000 [ 82%]  (Sampling)
## Chain 2 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 3 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 4 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 2 Iteration: 4900 / 5000 [ 98%]  (Sampling)
## Chain 3 Iteration: 4900 / 5000 [ 98%]  (Sampling)
## Chain 4 Iteration: 4300 / 5000 [ 86%]  (Sampling)
```

```
## Chain 2 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 2 finished in 52.4 seconds.
## Chain 3 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 3 finished in 53.0 seconds.
## Chain 4 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 4 Iteration: 4500 / 5000 [ 90%]  (Sampling)
## Chain 4 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 4 Iteration: 4700 / 5000 [ 94%]  (Sampling)
## Chain 4 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 4 Iteration: 4900 / 5000 [ 98%]  (Sampling)
## Chain 4 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 4 finished in 61.1 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 52.6 seconds.
## Total execution time: 61.2 seconds.
```

```
# print number of rhats above 1.1
s <- fit_refresh_lower$summary()
s$rhat <- as.numeric(s$rhat)

subset(s, rhat > 1.1 & grepl("^(pi|theta)\\b", variable))[, c("variable","rhat","ess_bulk","ess_tail")]
```

```
## # A tibble: 0 x 4
## # i 4 variables: variable <chr>, rhat <dbl>, ess_bulk <dbl>, ess_tail <dbl>
```

```
# print string if no rhats above 1.1
if (nrow(subset(s, rhat > 1.1 & grepl("^(pi|theta)\\b", variable))) == 0) {
  cat("No R-hat values above 1.1\n")
}
```

```
## No R-hat values above 1.1
```

```
fit_refresh_higher <- mod$sample(
  data = stan_data,
  seed = 1234,
  chains = 4,
  parallel_chains = 4,
  threads_per_chain = 1,
  iter_warmup = 1000,
  iter_sampling = 4000,
  refresh = 400,
  adapt_delta = 0.99
)
```

```
## Running MCMC with 4 parallel chains, with 1 thread(s) per chain...
##
## Chain 1 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 2 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 3 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 4 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 1 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 3 Iteration:  400 / 5000 [  8%]  (Warmup)
```

```
## Chain 4 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 2 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 1 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 3 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 2 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 4 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 1 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 3 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 2 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 4 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 1 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 3 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 2 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 4 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 1 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 3 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 2 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 1 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 4 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 1 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 3 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 2 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 4 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 1 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 3 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 2 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 1 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 4 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 3 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 2 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 1 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 4 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 3 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 2 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 1 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 3 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 1 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 4 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 2 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 1 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 1 finished in 43.5 seconds.
## Chain 3 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 2 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 4 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 3 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 2 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 4 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 3 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 3 finished in 53.2 seconds.
## Chain 2 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 2 finished in 53.7 seconds.
## Chain 4 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 4 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 4 finished in 60.8 seconds.
```

```
##
## All 4 chains finished successfully.
## Mean chain execution time: 52.8 seconds.
## Total execution time: 61.0 seconds.
```

```
# print number of rhats above 1.1
s <- fit_refresh_higher$summary()
s$rhat <- as.numeric(s$rhat)

subset(s, rhat > 1.1 & grepl("^(pi|theta)\\b", variable))[, c("variable","rhat","ess_bulk","ess_tail")]
```

```
## # A tibble: 0 x 4
## # i 4 variables: variable <chr>, rhat <dbl>, ess_bulk <dbl>, ess_tail <dbl>
```

```
# print string if no rhats above 1.1
if (nrow(subset(s, rhat > 1.1 & grepl("^(pi|theta)\\b", variable))) == 0) {
  cat("No R-hat values above 1.1\n")
}
```

```
## No R-hat values above 1.1
```

```
fit_adapt_delta_lower <- mod$sample(
  data = stan_data,
  seed = 1234,
  chains = 4,
  parallel_chains = 4,
  threads_per_chain = 1,
  iter_warmup = 1000,
  iter_sampling = 4000,
  refresh = 200,
  adapt_delta = 0.9
)
```

```
## Running MCMC with 4 parallel chains, with 1 thread(s) per chain...
##
## Chain 1 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 2 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 3 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 4 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 1 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 2 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 3 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 4 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 1 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 3 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 2 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 4 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 1 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 2 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 3 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 4 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 1 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 2 Iteration:  800 / 5000 [ 16%]  (Warmup)
```

```
## Chain 3 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 4 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 1 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 2 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 1 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 2 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 3 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 3 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 4 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 4 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 2 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 1 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 3 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 4 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 2 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 3 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 4 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 1 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 2 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 3 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 4 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 1 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 2 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 3 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 4 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 2 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 1 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 3 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 4 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 2 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 1 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 3 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 4 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 2 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 3 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 4 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 1 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 2 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 3 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 4 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 1 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 2 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 3 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 4 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 2 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 1 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 3 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 4 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 2 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 1 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 3 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 4 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 2 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 3 Iteration: 3400 / 5000 [ 68%]  (Sampling)
```

```
## Chain 4 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 1 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 2 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 3 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 4 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 2 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 1 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 4 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 3 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 2 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 1 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 4 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 2 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 3 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 4 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 1 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 2 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 3 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 4 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 2 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 3 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 1 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 4 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 2 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 3 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 1 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 4 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 2 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 3 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 2 finished in 21.3 seconds.
## Chain 1 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 4 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 4 finished in 22.0 seconds.
## Chain 3 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 3 finished in 22.2 seconds.
## Chain 1 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 1 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 1 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 1 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 1 finished in 26.4 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 23.0 seconds.
## Total execution time: 26.5 seconds.
```

```
# print number of rhats above 1.1
s <- fit_adapt_delta_lower$summary()
s$rhat <- as.numeric(s$rhat)

subset(s, rhat > 1.1 & grepl("^(pi|theta)\\b", variable))[, c("variable","rhat","ess_bulk","ess_tail")]
```

```
## # A tibble: 0 x 4
## # i 4 variables: variable <chr>, rhat <dbl>, ess_bulk <dbl>, ess_tail <dbl>
```

```r
# print string if no rhats above 1.1
if (nrow(subset(s, rhat > 1.1 & grepl("^(pi|theta)\\b", variable))) == 0) {
  cat("No R-hat values above 1.1\n")
}
```

```
## No R-hat values above 1.1
```

```r
fit_adapt_delta_higher <- mod$sample(
  data = stan_data,
  seed = 1234,
  chains = 4,
  parallel_chains = 4,
  threads_per_chain = 1,
  iter_warmup = 1000,
  iter_sampling = 4000,
  refresh = 200,
  adapt_delta = 0.999
)
```

```
## Running MCMC with 4 parallel chains, with 1 thread(s) per chain...
##
## Chain 1 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 2 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 3 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 4 Iteration:    1 / 5000 [  0%]  (Warmup)
## Chain 1 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 2 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 3 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 1 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 4 Iteration:  200 / 5000 [  4%]  (Warmup)
## Chain 3 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 1 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 2 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 4 Iteration:  400 / 5000 [  8%]  (Warmup)
## Chain 3 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 1 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 2 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 3 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 2 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 4 Iteration:  600 / 5000 [ 12%]  (Warmup)
## Chain 1 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 1 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 3 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 3 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 4 Iteration:  800 / 5000 [ 16%]  (Warmup)
## Chain 1 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 2 Iteration: 1000 / 5000 [ 20%]  (Warmup)
## Chain 2 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 3 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 1 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 4 Iteration: 1000 / 5000 [ 20%]  (Warmup)
```

```
## Chain 4 Iteration: 1001 / 5000 [ 20%]  (Sampling)
## Chain 1 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 2 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 3 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 1 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 3 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 4 Iteration: 1200 / 5000 [ 24%]  (Sampling)
## Chain 1 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 2 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 3 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 1 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 4 Iteration: 1400 / 5000 [ 28%]  (Sampling)
## Chain 3 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 2 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 1 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 3 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 1 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 4 Iteration: 1600 / 5000 [ 32%]  (Sampling)
## Chain 2 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 1 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 3 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 1 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 3 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 2 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 4 Iteration: 1800 / 5000 [ 36%]  (Sampling)
## Chain 1 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 3 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 1 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 2 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 3 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 4 Iteration: 2000 / 5000 [ 40%]  (Sampling)
## Chain 1 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 3 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 1 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 2 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 4 Iteration: 2200 / 5000 [ 44%]  (Sampling)
## Chain 1 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 3 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 1 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 3 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 2 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 4 Iteration: 2400 / 5000 [ 48%]  (Sampling)
## Chain 1 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 3 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 1 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 2 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 3 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 1 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 4 Iteration: 2600 / 5000 [ 52%]  (Sampling)
## Chain 3 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 1 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 1 finished in 57.3 seconds.
## Chain 2 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 3 Iteration: 4400 / 5000 [ 88%]  (Sampling)
```

```
## Chain 4 Iteration: 2800 / 5000 [ 56%]  (Sampling)
## Chain 2 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 3 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 4 Iteration: 3000 / 5000 [ 60%]  (Sampling)
## Chain 3 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 2 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 3 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 3 finished in 68.0 seconds.
## Chain 4 Iteration: 3200 / 5000 [ 64%]  (Sampling)
## Chain 2 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 4 Iteration: 3400 / 5000 [ 68%]  (Sampling)
## Chain 2 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 2 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 4 Iteration: 3600 / 5000 [ 72%]  (Sampling)
## Chain 2 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 4 Iteration: 3800 / 5000 [ 76%]  (Sampling)
## Chain 2 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 4 Iteration: 4000 / 5000 [ 80%]  (Sampling)
## Chain 2 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 4 Iteration: 4200 / 5000 [ 84%]  (Sampling)
## Chain 2 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 4 Iteration: 4400 / 5000 [ 88%]  (Sampling)
## Chain 2 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 2 finished in 101.2 seconds.
## Chain 4 Iteration: 4600 / 5000 [ 92%]  (Sampling)
## Chain 4 Iteration: 4800 / 5000 [ 96%]  (Sampling)
## Chain 4 Iteration: 5000 / 5000 [100%]  (Sampling)
## Chain 4 finished in 113.2 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 84.9 seconds.
## Total execution time: 113.3 seconds.
```

```
# print number of rhats above 1.1
s <- fit_adapt_delta_higher$summary()
s$rhat <- as.numeric(s$rhat)

subset(s, rhat > 1.1 & grepl("^(pi|theta)\\b", variable))[, c("variable","rhat","ess_bulk","ess_tail")]
```

```
## # A tibble: 0 x 4
## # i 4 variables: variable <chr>, rhat <dbl>, ess_bulk <dbl>, ess_tail <dbl>
```

```
# print string if no rhats above 1.1
if (nrow(subset(s, rhat > 1.1 & grepl("^(pi|theta)\\b", variable))) == 0) {
  cat("No R-hat values above 1.1\n")
}
```

```
## No R-hat values above 1.1
```

```
if (interactive() && is.null(knitr::opts_knit$get("rmarkdown.pandoc.to"))) {
  library(shinystan)
  launch_shinystan(fit_basis)
}

if (interactive() && is.null(knitr::opts_knit$get("rmarkdown.pandoc.to"))) {
  library(shinystan)
```

```
  launch_shinystan(fit_adapt_delta_lower)
}
```

Robustness check - Are there differences in the results if one changes convergence settings?

```
library(posterior)
names(summarise_draws(as_draws_df(fit_basis$draws("pi")), mean, sd, ~quantile(.x, c(.05, .95))))
```

```
## [1] "variable" "mean"     "sd"       "5%"       "95%"
```

```
compare_param <- function(fit1, fit2, var) {
  d1 <- as_draws_df(fit1$draws(var))
  d2 <- as_draws_df(fit2$draws(var))

  s1 <- summarise_draws(d1, mean, sd, ~quantile(.x, c(.05, .95), na.rm = TRUE))
  s2 <- summarise_draws(d2, mean, sd, ~quantile(.x, c(.05, .95), na.rm = TRUE))

  out <- s1[, c("variable", "mean", "sd", "5%", "95%")]
  out$mean_2 <- s2$mean
  out$sd_2   <- s2$sd
  out$`5%_2` <- s2$`5%`
  out$`95%_2` <- s2$`95%`
  out$diff_mean <- out$mean_2 - out$mean
  out
}


# 1) Compare parameters
pi_cmp    <- compare_param(fit_basis, fit_adapt_delta_lower, "pi")
theta_cmp <- compare_param(fit_basis, fit_adapt_delta_lower, "theta")

#2) Show largest differences
pi_cmp[order(abs(pi_cmp$diff_mean), decreasing = TRUE), ]
```

```
## # A tibble: 2 x 10
##   variable  mean     sd  `5%` `95%` mean_2   sd_2 `5%_2` `95%_2` diff_mean
##   <chr>    <dbl>  <dbl> <dbl> <dbl>  <dbl>  <dbl>  <dbl>   <dbl>     <dbl>
## 1 pi[1]    0.726 0.0240 0.685 0.765  0.726 0.0235  0.686   0.764 -0.000236
## 2 pi[2]    0.274 0.0240 0.235 0.315  0.274 0.0235  0.236   0.314  0.000236
```

```
theta_cmp[order(abs(theta_cmp$diff_mean), decreasing = TRUE), ][1:20, ]
```

```
## # A tibble: 20 x 10
##    variable        mean       sd     `5%`   `95%`  mean_2     sd_2   `5%_2` `95%_2`  diff_mean
##    <chr>          <dbl>    <dbl>    <dbl>   <dbl>   <dbl>    <dbl>    <dbl>   <dbl>      <dbl>
##  1 theta[3,1,2]  0.0471  0.0131   0.0276   0.0702  0.0476  0.0136   0.0275   0.0718  0.000442
##  2 theta[3,1,1]  0.953   0.0131   0.930    0.972   0.952   0.0136   0.928    0.973  -0.000442
##  3 theta[2,1,2]  0.0134  0.00838  0.00232  0.0293  0.0137  0.00849  0.00244  0.0298  0.000218
##  4 theta[2,1,1]  0.987   0.00838  0.971    0.998   0.986   0.00849  0.970    0.998  -0.000218
##  5 theta[2,2,1]  0.0205  0.0166   0.00185  0.0529  0.0204  0.0170   0.00165  0.0532 -0.000144
##  6 theta[2,2,2]  0.979   0.0166   0.947    0.998   0.980   0.0170   0.947    0.998   0.000144
##  7 theta[1,1,1]  0.953   0.0135   0.929    0.973   0.953   0.0133   0.929    0.973  -0.000135
```

```
##  8 theta[1,1,2]  0.0469  0.0135   0.0268   0.0708  0.0470  0.0133   0.0270   0.0710  0.000135
##  9 theta[1,2,1]  0.0676  0.0274   0.0286   0.118   0.0677  0.0271   0.0292   0.117   0.0000876
## 10 theta[1,2,2]  0.932   0.0274   0.882    0.971   0.932   0.0271   0.883    0.971  -0.0000876
## 11 theta[3,2,1]  0.130   0.0350   0.0759   0.191   0.129   0.0351   0.0760   0.191  -0.0000544
## 12 theta[3,2,2]  0.870   0.0350   0.809    0.924   0.871   0.0351   0.809    0.924   0.0000544
## 13 <NA>          NA      NA       NA       NA      NA      NA       NA       NA      NA
## 14 <NA>          NA      NA       NA       NA      NA      NA       NA       NA      NA
## 15 <NA>          NA      NA       NA       NA      NA      NA       NA       NA      NA
## 16 <NA>          NA      NA       NA       NA      NA      NA       NA       NA      NA
## 17 <NA>          NA      NA       NA       NA      NA      NA       NA       NA      NA
## 18 <NA>          NA      NA       NA       NA      NA      NA       NA       NA      NA
## 19 <NA>          NA      NA       NA       NA      NA      NA       NA       NA      NA
## 20 <NA>          NA      NA       NA       NA      NA      NA       NA       NA      NA
```

```
# show only differences larger than 0.001
subset(theta_cmp, abs(diff_mean) > 1e-3)
```

```
## # A tibble: 0 x 10
## # i 10 variables: variable <chr>, mean <dbl>, sd <dbl>, 5% <dbl>, 95% <dbl>, mean_2 <dbl>, sd_2 <dbl
## #   5%_2 <dbl>, 95%_2 <dbl>, diff_mean <dbl>
```

```
# no content

theta_cmp$overlap_5_95 <- pmin(theta_cmp$`95%`, theta_cmp$`95%_2`) -
  pmax(theta_cmp$`5%`,  theta_cmp$`5%_2`)

# negative values mean no overlap
theta_cmp[order(theta_cmp$overlap_5_95), ][1:20, ]
```

```
## # A tibble: 20 x 11
##    variable        mean       sd     `5%`   `95%`  mean_2     sd_2   `5%_2` `95%_2`  diff_mean overla
##    <chr>          <dbl>    <dbl>    <dbl>   <dbl>   <dbl>    <dbl>    <dbl>   <dbl>      <dbl>
##  1 theta[2,1,1]  0.987   0.00838  0.971    0.998   0.986   0.00849  0.970    0.998  -0.000218
##  2 theta[2,1,2]  0.0134  0.00838  0.00232  0.0293  0.0137  0.00849  0.00244  0.0298  0.000218
##  3 theta[3,1,1]  0.953   0.0131   0.930    0.972   0.952   0.0136   0.928    0.973  -0.000442
##  4 theta[3,1,2]  0.0471  0.0131   0.0276   0.0702  0.0476  0.0136   0.0275   0.0718  0.000442
##  5 theta[1,1,2]  0.0469  0.0135   0.0268   0.0708  0.0470  0.0133   0.0270   0.0710  0.000135
##  6 theta[1,1,1]  0.953   0.0135   0.929    0.973   0.953   0.0133   0.929    0.973  -0.000135
##  7 theta[2,2,2]  0.979   0.0166   0.947    0.998   0.980   0.0170   0.947    0.998   0.000144
##  8 theta[2,2,1]  0.0205  0.0166   0.00185  0.0529  0.0204  0.0170   0.00165  0.0532 -0.000144
##  9 theta[1,2,1]  0.0676  0.0274   0.0286   0.118   0.0677  0.0271   0.0292   0.117   0.0000876
## 10 theta[1,2,2]  0.932   0.0274   0.882    0.971   0.932   0.0271   0.883    0.971  -0.0000876
## 11 theta[3,2,2]  0.870   0.0350   0.809    0.924   0.871   0.0351   0.809    0.924   0.0000544
## 12 theta[3,2,1]  0.130   0.0350   0.0759   0.191   0.129   0.0351   0.0760   0.191  -0.0000544
## 13 <NA>          NA      NA       NA       NA      NA      NA       NA       NA      NA
## 14 <NA>          NA      NA       NA       NA      NA      NA       NA       NA      NA
## 15 <NA>          NA      NA       NA       NA      NA      NA       NA       NA      NA
## 16 <NA>          NA      NA       NA       NA      NA      NA       NA       NA      NA
## 17 <NA>          NA      NA       NA       NA      NA      NA       NA       NA      NA
## 18 <NA>          NA      NA       NA       NA      NA      NA       NA       NA      NA
## 19 <NA>          NA      NA       NA       NA      NA      NA       NA       NA      NA
## 20 <NA>          NA      NA       NA       NA      NA      NA       NA       NA      NA
```

```
theta_cmp$diff_over_sd <- abs(theta_cmp$diff_mean) / theta_cmp$sd
theta_cmp[order(theta_cmp$diff_over_sd, decreasing = TRUE), ][1:20, ]
```

```
## # A tibble: 20 x 12
##    variable        mean       sd     `5%`   `95%`  mean_2     sd_2   `5%_2` `95%_2`  diff_mean overla
##    <chr>          <dbl>    <dbl>    <dbl>   <dbl>   <dbl>    <dbl>    <dbl>   <dbl>      <dbl>
##  1 theta[3,1,2]  0.0471  0.0131   0.0276   0.0702  0.0476  0.0136   0.0275   0.0718  0.000442
##  2 theta[3,1,1]  0.953   0.0131   0.930    0.972   0.952   0.0136   0.928    0.973  -0.000442
##  3 theta[2,1,2]  0.0134  0.00838  0.00232  0.0293  0.0137  0.00849  0.00244  0.0298  0.000218
##  4 theta[2,1,1]  0.987   0.00838  0.971    0.998   0.986   0.00849  0.970    0.998  -0.000218
##  5 theta[1,1,1]  0.953   0.0135   0.929    0.973   0.953   0.0133   0.929    0.973  -0.000135
##  6 theta[1,1,2]  0.0469  0.0135   0.0268   0.0708  0.0470  0.0133   0.0270   0.0710  0.000135
##  7 theta[2,2,1]  0.0205  0.0166   0.00185  0.0529  0.0204  0.0170   0.00165  0.0532 -0.000144
##  8 theta[2,2,2]  0.979   0.0166   0.947    0.998   0.980   0.0170   0.947    0.998   0.000144
##  9 theta[1,2,1]  0.0676  0.0274   0.0286   0.118   0.0677  0.0271   0.0292   0.117   0.0000876
## 10 theta[1,2,2]  0.932   0.0274   0.882    0.971   0.932   0.0271   0.883    0.971  -0.0000876
## 11 theta[3,2,1]  0.130   0.0350   0.0759   0.191   0.129   0.0351   0.0760   0.191  -0.0000544
## 12 theta[3,2,2]  0.870   0.0350   0.809    0.924   0.871   0.0351   0.809    0.924   0.0000544
## 13 <NA>          NA      NA       NA       NA      NA      NA       NA       NA      NA
## 14 <NA>          NA      NA       NA       NA      NA      NA       NA       NA      NA
## 15 <NA>          NA      NA       NA       NA      NA      NA       NA       NA      NA
## 16 <NA>          NA      NA       NA       NA      NA      NA       NA       NA      NA
## 17 <NA>          NA      NA       NA       NA      NA      NA       NA       NA      NA
## 18 <NA>          NA      NA       NA       NA      NA      NA       NA       NA      NA
## 19 <NA>          NA      NA       NA       NA      NA      NA       NA       NA      NA
## 20 <NA>          NA      NA       NA       NA      NA      NA       NA       NA      NA
## # i 1 more variable: diff_over_sd <dbl>
```

```r
# library(dplyr)
# library(posterior)
# library(ggplot2)
# library(reshape2)
#
# out_dir <- "img/model_results/isco/two_categories"
# dir.create(out_dir, recursive = TRUE, showWarnings = FALSE)
#
# save_plot <- function(p, filename, width = 7, height = 5, dpi = 600) {
#   ggsave(
#     filename = file.path(out_dir, filename),
#     plot = p,
#     width = width,
#     height = height,
#     dpi = dpi
#   )
# }
#
# # ------------------------------------------------------------
# # 0) Basic diagnostics for key parameters
# # ------------------------------------------------------------
# diag_tbl <- fit_basis$summary(variables = c("pi", "theta")) %>%
#   dplyr::select(variable, rhat, ess_bulk, ess_tail)
#
# diag_bad <- diag_tbl %>%
#   dplyr::filter(is.finite(rhat), rhat > 1.01) %>%
#   dplyr::arrange(desc(rhat))
#
# diag_bad
```

```
#
# # Divergences, treedepth hits, if available (cmdstanr)
# div_info <- tryCatch(fit_basis$diagnostic_summary(), error = function(e) NULL)
# if (is.null(div_info)) {
#   div_info <- tryCatch(fit_basis$cmdstan_diagnose(), error = function(e) NULL)
# }
# div_info
#
# # ------------------------------------------------------------
# # 1) Confusion matrices (theta), posterior mean and uncertainty
# # ------------------------------------------------------------
# theta_draws_mat <- fit_basis$draws("theta", format = "draws_matrix")
# theta_mean_vec  <- colMeans(theta_draws_mat)
#
# # Expectation: theta has dimensions [J rater, K true, K given]
# theta_mean_arr <- array(
#   theta_mean_vec,
#   dim = c(J, K, K),
#   dimnames = list(
#     rater = c("human1", "human2", "tool"),
#     true  = isco_levels,
#     given = isco_levels
#   )
# )
#
# theta_human1_mean <- theta_mean_arr["human1", , ]
# theta_human2_mean <- theta_mean_arr["human2", , ]
# theta_tool_mean   <- theta_mean_arr["tool",   , ]
#
# theta_human1_mean
# rowSums(theta_human1_mean)
#
# theta_human2_mean
# rowSums(theta_human2_mean)
#
# theta_tool_mean
# rowSums(theta_tool_mean)
#
# plot_theta <- function(mat, title_txt) {
#   long <- reshape2::melt(mat, varnames = c("True", "Given"), value.name = "Prob")
#   ggplot(long, aes(x = Given, y = True, fill = Prob)) +
#     geom_tile(color = "white") +
#     scale_fill_gradient(low = "white", high = "darkgreen") +
#     theme_light() +
#     labs(
#       title = title_txt,
#       x = "Observed class",
#       y = "True class (latent)"
#     ) +
#     theme(axis.text.x = element_text(angle = 45, hjust = 1))
# }
#
# p_theta_h1_mean <- plot_theta(theta_human1_mean, "Confusion matrix, human 1 (posterior mean)")
```

```
# p_theta_h2_mean <- plot_theta(theta_human2_mean, "Confusion matrix, human 2 (posterior mean)")
# p_theta_tool_mean <- plot_theta(theta_tool_mean, "Confusion matrix, tool (posterior mean)")
#
# save_plot(p_theta_h1_mean, "theta_human1_mean.png")
# save_plot(p_theta_h2_mean, "theta_human2_mean.png")
# save_plot(p_theta_tool_mean, "theta_tool_mean.png")
#
# p_theta_h1_mean
# p_theta_h2_mean
# p_theta_tool_mean
#
# # Credible intervals for all theta entries
# theta_summ <- summarise_draws(
#   as_draws_df(fit_basis$draws("theta")),
#   mean, sd,
#   ~quantile(.x, probs = c(`5%` = 0.05, `50%` = 0.50, `95%` = 0.95), na.rm = TRUE)
# )
# theta_summ
#
# # Posterior SD heatmaps for theta (uncertainty by cell)
# theta_sd_vec <- apply(theta_draws_mat, 2, sd)
# theta_sd_arr <- array(
#   theta_sd_vec,
#   dim = c(J, K, K),
#   dimnames = list(
#     rater = c("human1", "human2", "tool"),
#     true  = isco_levels,
#     given = isco_levels
#   )
# )
#
# p_theta_h1_sd <- plot_theta(theta_sd_arr["human1", , ], "Confusion matrix, human 1 (posterior SD)")
# p_theta_h2_sd <- plot_theta(theta_sd_arr["human2", , ], "Confusion matrix, human 2 (posterior SD)")
# p_theta_tool_sd <- plot_theta(theta_sd_arr["tool", , ], "Confusion matrix, tool (posterior SD)")
#
# save_plot(p_theta_h1_sd, "theta_human1_sd.png")
# save_plot(p_theta_h2_sd, "theta_human2_sd.png")
# save_plot(p_theta_tool_sd, "theta_tool_sd.png")
#
# p_theta_h1_sd
# p_theta_h2_sd
# p_theta_tool_sd
#
# # ------------------------------------------------------------
# # 2) Class prevalence (pi), posterior mean and uncertainty
# # ------------------------------------------------------------
# pi_summ <- summarise_draws(
#   as_draws_df(fit_basis$draws("pi")),
#   mean, sd,
#   ~quantile(.x, probs = c(`5%` = 0.05, `50%` = 0.50, `95%` = 0.95), na.rm = TRUE)
# )
# pi_summ
#
```

```
# pi_means <- colMeans(as_draws_matrix(fit_basis$draws("pi")))
# pi_df <- data.frame(
#   class = isco_levels,
#   prevalence = as.numeric(pi_means)
# )
#
# p_pi_bar <- ggplot(pi_df, aes(x = class, y = prevalence)) +
#   geom_bar(stat = "identity", fill = "#02626f") +
#   theme_light() +
#   labs(
#     title = "Posterior mean of class prevalence (p)",
#     x = "Class",
#     y = "Prevalence"
#   ) +
#   theme(axis.text.x = element_text(angle = 45, hjust = 1))
#
# save_plot(p_pi_bar, "pi_mean_bar.png", width = 6, height = 4)
# p_pi_bar
#
# # ------------------------------------------------------------
# # 2b) Rater level scalar summaries from theta
# # ------------------------------------------------------------
# pi_mean_vec <- as.numeric(pi_means)
# names(pi_mean_vec) <- isco_levels
#
# weighted_diag_mass <- function(mat, w) sum(diag(mat) * w)
# weighted_offdiag_mass <- function(mat, w) sum((1 - diag(mat)) * w)
#
# rater_summary <- data.frame(
#   rater = c("human1", "human2", "tool"),
#   weighted_diag_mass = c(
#     weighted_diag_mass(theta_human1_mean, pi_mean_vec),
#     weighted_diag_mass(theta_human2_mean, pi_mean_vec),
#     weighted_diag_mass(theta_tool_mean,   pi_mean_vec)
#   ),
#   weighted_offdiag_mass = c(
#     weighted_offdiag_mass(theta_human1_mean, pi_mean_vec),
#     weighted_offdiag_mass(theta_human2_mean, pi_mean_vec),
#     weighted_offdiag_mass(theta_tool_mean,   pi_mean_vec)
#   )
# )
# rater_summary
#
# p_rater_diag <- ggplot(rater_summary, aes(x = rater, y = weighted_diag_mass)) +
#   geom_bar(stat = "identity", fill = "#02626f") +
#   theme_light() +
#   labs(
#     title = "Rater summary: Weighted diagonal mass",
#     x = "Rater",
#     y = "Sum_j pi_j * theta_jj"
#   )
#
# save_plot(p_rater_diag, "rater_weighted_diag_mass.png", width = 6, height = 4)
```

```
# p_rater_diag
#
# # ------------------------------------------------------------
# # 3) Posterior class probabilities per item (q_z)
# # ------------------------------------------------------------
# qz_draws_mat <- fit_basis$draws("q_z", format = "draws_matrix")
# qz_mean_vec  <- colMeans(qz_draws_mat)
#
# qz_mean <- matrix(
#   qz_mean_vec,
#   nrow = I, ncol = K,
#   dimnames = list(item = item_ids, class = isco_levels)
# )
#
# qz_df <- as.data.frame(qz_mean)
# p_cols <- paste0("p_", isco_levels)
# colnames(qz_df) <- p_cols
# qz_df$item_id <- item_ids
#
# # MAP label
# qz_df$map_class_idx   <- max.col(qz_df[p_cols], ties.method = "first")
# qz_df$map_class_label <- isco_levels[qz_df$map_class_idx]
#
# # Uncertainty measures
# eps <- 1e-12
# qz_df$max_p <- apply(qz_df[p_cols], 1, max)
#
# qz_df$entropy <- -rowSums(as.matrix(qz_df[p_cols]) * log(as.matrix(qz_df[p_cols]) + eps))
#
# qz_df$margin <- apply(qz_df[p_cols], 1, function(p) {
#   ps <- sort(p, decreasing = TRUE)
#   ps[1] - ps[2]
# })
#
# # Credible set per item, smallest set reaching cumulative posterior mass >= level
# credible_set <- function(p, classes, level = 0.95) {
#   o <- order(p, decreasing = TRUE)
#   cum <- cumsum(p[o])
#   m <- which(cum >= level)[1]
#   classes[o[seq_len(m)]]
# }
# qz_df$cs95 <- apply(qz_df[p_cols], 1, function(x) credible_set(x, isco_levels, level = 0.95))
#
# table(lengths(qz_df$cs95))
#
# # Plots of uncertainty and confidence
# p_maxp_hist <- ggplot(qz_df, aes(x = max_p)) +
#   geom_histogram(binwidth = 0.025, fill = "#333333", color = "white", size = 0.1) +
#   theme_light() +
#   labs(x = "Posterior max probability", y = "Count",
#        title = "Model confidence per item")
#
# p_entropy_hist <- ggplot(qz_df, aes(x = entropy)) +
```

```
#   geom_histogram(binwidth = 0.025, fill = "#333333", color = "white", size = 0.1) +
#   theme_light() +
#   labs(x = "Posterior entropy", y = "Count",
#        title = "Item uncertainty (entropy)")
#
# p_margin_hist <- ggplot(qz_df, aes(x = margin)) +
#   geom_histogram(binwidth = 0.025, fill = "#333333", color = "white", size = 0.1) +
#   theme_light() +
#   labs(x = "Posterior margin (top1 minus top2)", y = "Count",
#        title = "Item uncertainty (margin)")
#
# save_plot(p_maxp_hist, "qz_maxp_hist.png", width = 6, height = 4)
# save_plot(p_entropy_hist, "qz_entropy_hist.png", width = 6, height = 4)
# save_plot(p_margin_hist, "qz_margin_hist.png", width = 6, height = 4)
#
# p_maxp_hist
# p_entropy_hist
# p_margin_hist
#
# # ------------------------------------------------------------
# # 3b) MAP distribution vs pi (sanity check)
# # ------------------------------------------------------------
# map_tab <- qz_df %>%
#   count(map_class_label, name = "n_map") %>%
#   mutate(map_share = n_map / sum(n_map)) %>%
#   rename(class = map_class_label)
#
# pi_tab <- data.frame(class = isco_levels, pi_mean = as.numeric(pi_means))
#
# map_vs_pi <- left_join(map_tab, pi_tab, by = "class")
# map_vs_pi
#
# p_map_vs_pi <- ggplot(map_vs_pi, aes(x = class)) +
#   geom_col(aes(y = map_share), fill = "#02626f") +
#   geom_point(aes(y = pi_mean), size = 2) +
#   theme_light() +
#   labs(
#     title = "MAP share (bars) vs posterior mean pi (points)",
#     x = "Class",
#     y = "Share / prevalence"
#   )
#
# save_plot(p_map_vs_pi, "map_share_vs_pi.png", width = 6, height = 4)
# p_map_vs_pi
#
# # ------------------------------------------------------------
# # 4) Merge predictions with original ratings, include uncertainty
# # ------------------------------------------------------------
# ratings <- isco_data %>%
#   dplyr::select(item_id, isco_human1, isco_human2, isco_tool)
#
# predictions <- qz_df %>%
#   dplyr::select(item_id, map_class_label, all_of(p_cols), max_p, entropy, margin, cs95)
```

```
#
# ratings_with_predictions <- ratings %>%
#   dplyr::left_join(predictions, by = "item_id") %>%
#   dplyr::rename(
#     predicted_class = map_class_label,
#     rating_human1 = isco_human1,
#     rating_human2 = isco_human2,
#     rating_tool   = isco_tool
#   ) %>%
#   dplyr::arrange(item_id)
#
# head(ratings_with_predictions, 30)
#
# # Most uncertain items by entropy, include posterior probabilities
# most_uncertain <- ratings_with_predictions %>%
#   dplyr::arrange(desc(entropy)) %>%
#   dplyr::select(item_id, predicted_class, entropy, margin, max_p, all_of(p_cols),
#                 rating_human1, rating_human2, rating_tool)
#
# head(most_uncertain, 20)
#
# # ------------------------------------------------------------
# # 5) Observed agreement, descriptive, not model based
# # ------------------------------------------------------------
# mean(isco_data$isco_human1 == isco_data$isco_tool, na.rm = TRUE)
# mean(isco_data$isco_human2 == isco_data$isco_tool, na.rm = TRUE)
#
# isco_data %>%
#   dplyr::count(isco_human1, isco_human2, isco_tool) %>%
#   dplyr::arrange(desc(n))
#
```

```
# # 5) Posterior summaries
# # Confusion matrices per rater (posterior mean)
# theta_draws_rvars <- fit_adapt_delta_lower$draws("theta")
# theta_rvars <- posterior::as_draws_rvars(theta_draws_rvars)
# theta_mean_rvar <- posterior::rvar_mean(theta_rvars$theta)
#
# # Get draws directly as a matrix
# theta_draws_mat <- fit_adapt_delta_lower$draws("theta", format = "draws_matrix")
#
# # Mean across all draws
# theta_mean_vec <- colMeans(theta_draws_mat)
#
# # Reshape into a 3D array (J x K x K)
# theta_mean_arr <- array(
#   theta_mean_vec,
#   dim = c(J, K, K),
#   dimnames = list(
#     rater = c("human1", "human2", "tool"),
#     true  = isco_levels,
#     given = isco_levels
#   )
# )
```

```
#
# # Example: Human 1 (rater 1)
# theta_human1_mean <- theta_mean_arr["human1", , ]
# theta_human1_mean
#
# rowSums(theta_human1_mean)
# # Should all be about 1
#
# # Same for rater 2 and tool
# theta_human2_mean <- theta_mean_arr["human2", , ]
# theta_tool_mean   <- theta_mean_arr["tool", , ]
#
# theta_human2_mean
# theta_tool_mean
#
# # Posterior class probabilities per item (with original labels)
# qz_draws_array <- fit_adapt_delta_lower$draws("q_z", format = "draws_array")    # [iter, chain, I, K]
# qz_draws_mat   <- fit_adapt_delta_lower$draws("q_z", format = "draws_matrix")
#
# # Mean across all draws
# qz_mean_vec <- colMeans(qz_draws_mat)
#
# # Reshape into an I x K matrix
# qz_mean <- matrix(
#   qz_mean_vec,
#   nrow = I, ncol = K,
#   dimnames = list(item = item_ids, class = isco_levels)
# )
#
# qz_df <- as.data.frame(qz_mean)
# colnames(qz_df) <- paste0("p_", isco_levels)
# qz_df$item_id <- item_ids
#
# # MAP, both as index and as original label
# qz_df$map_class_idx   <- max.col(qz_df[paste0("p_", isco_levels)], ties.method = "first")
# qz_df$map_class_label <- isco_levels[qz_df$map_class_idx]
#
# # Join back to items
# result_items <- data.frame(item_id = item_ids) %>%
#   dplyr::left_join(qz_df, by = "item_id")
#
# head(result_items)
# print(head(result_items, 20))
#
# isco_data %>%
#   dplyr::count(isco_human1, isco_human2, isco_tool) %>%
#   dplyr::arrange(desc(n))
#
# mean(isco_data$isco_human1 == isco_data$isco_tool, na.rm = TRUE)
# mean(isco_data$isco_human2 == isco_data$isco_tool, na.rm = TRUE)
#
# library(ggplot2)
# qz_df$max_p <- apply(qz_df[paste0("p_", isco_levels)], 1, max)
```

```
#
# ggplot(qz_df, aes(x = max_p)) +
#   geom_histogram(binwidth = 0.025, fill = "#333333", color = "white", size = 0.1) +
#   labs(x = "Posterior max probability", y = "Count",
#        title = "Model confidence per item")
#
# # Output from already executed code
#
# rowSums(theta_human1_mean)
# # Should all be about 1
#
# library(reshape2)
# theta_human1_long <- melt(theta_human1_mean, varnames = c("True", "Given"), value.name = "Prob")
#
# ggplot(theta_human1_long, aes(x = Given, y = True, fill = Prob)) +
#   geom_tile(color = "white") +
#   scale_fill_gradient(low = "white", high = "darkgreen") +
#   theme_light() +
#   labs(
#     title = "Confusion matrix (Human rater 1)",
#     x = "Observed class",
#     y = "True class (latent)"
#   ) +
#   theme(axis.text.x = element_text(angle = 45, hjust = 1))
#
# library(posterior)
# pi_means <- colMeans(as_draws_matrix(fit_adapt_delta_lower$draws("pi")))
# pi_means
#
# isco_levels[1]
#
# # Visualize pi_means in a bar chart
# pi_df <- data.frame(
#   class = isco_levels,
#   prevalence = pi_means
# )
#
# ggplot(pi_df, aes(x = class, y = prevalence)) +
#   geom_bar(stat = "identity", fill = "#02626f") +
#   theme_light() +
#   labs(
#     title = "Posterior mean of class prevalence (p)",
#     x = "Class",
#     y = "Prevalence"
#   ) +
#   theme(axis.text.x = element_text(angle = 45, hjust = 1))
#
# # Merge ratings and predicted class
#
# # 1) Get the original rater labels again
# ratings <- isco_data %>%
#   dplyr::select(item_id, isco_human1, isco_human2, isco_tool)
#
```

```
# # 2) Combine with posterior class assignments
# predictions <- qz_df %>%
#   dplyr::select(item_id, map_class_label, starts_with("p_"))
#
# # 3) Merge
# ratings_with_predictions <- ratings %>%
#   dplyr::left_join(predictions, by = "item_id") %>%
#   dplyr::arrange(item_id)
#
# # 4) Readable column names
# ratings_with_predictions <- ratings_with_predictions %>%
#   dplyr::rename(
#     predicted_class = map_class_label,
#     rating_human1 = isco_human1,
#     rating_human2 = isco_human2,
#     rating_tool   = isco_tool
#   )
#
# # 5) Example output
# head(ratings_with_predictions, 30)
```